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Mmod functions

Synopsis

The functions prefixed with Mmod make up the Geometric Model Finder module, also referred to as the Model Finder module. The Geometric Model Finder module finds occurrences of models, or sets of image characteristics, in a target using their edge-based geometric features (geometric features from extracted edges). The module can help solve machine vision problems, such as alignment, measurement, and inspection. The Model Finder control also provides complete support for camera calibration. Searches are performed in the calibrated real-world such that, even without physically correcting your images, complex distortions do not affect performance, and results can be returned in real-world units.

Functions

MmodAlloc
MmodAllocResult
MmodControl
MmodDefine
MmodDefineFromFile
MmodDraw
MmodFind
MmodFree
MmodGetResult
MmodInquire
MmodMask
MmodPreprocess
MmodRestore
MmodSave
MmodStream

MmodAlloc

Synopsis

Allocate a Model Finder context.

Syntax

```
AIL_ID MmodAlloc(
    AIL_ID SysId,              //in
    AIL_INT64 ModelFinderType, //in
    AIL_INT64 ControlFlag,     //in
    AIL_ID *ContextIdPtr       //out
)
```

Description

This function allocates a Model Finder context on the specified system. A Model Finder context contains all the information necessary to perform an MmodFind() search, including global search settings, and the individual model(s) to locate.

Define and add models to the Model Finder context using MmodDefine(). You must add at least one model to the context regardless of the context type. Some context types accept only one model.

The Model Finder context and individual model search settings can be adjusted using MmodControl().

When the Model Finder context is no longer required, release it using MmodFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the Model Finder context's lifetime and you must not manually free it.

Parameters

SysId INQ

Specifies the system on which to allocate the context. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ModelFinderType INQ

Specifies the type of Model Finder context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the type of context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GEOMETRIC	Specifies that the Model Finder context uses a general geometric search algorithm. This algorithm uses geometric features to locate user-specified model(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_GEOMETRIC_CONTROLLED	Specifies that the Model Finder context uses a controlled geometric search algorithm. When there is a lot of geometric complexity, this search method is often faster and more robust than M_GEOMETRIC. The search method is especially fast when the angle between the occurrence and the nominal angle of the model (M_ANGLE) is small, although searching within an angular range is supported. This search method is recommended when there are only small differences in scale between the occurrence and the nominal scale of the model (M_SCALE); searching through a range of scales is not supported (M_SEARCH_SCALE_RANGE is disabled by default and cannot be set to M_ENABLE). You cannot get or draw the target edges for the entire target (M_GENERAL); you can only get or draw them in the region of an occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

		c		e	f	g	a		i		j	k	l	m	n	b
M_SHAPE_CIRCLE	Specifies that the Model Finder context uses a circular model search algorithm. This context only supports M_CIRCLE type models, and only supports one model per context. The model cannot be calibrated. If the target is associated with a camera calibration context, the model will be interpreted in world units; conversely, if the target does not have a camera calibration context associated with it, the model will be interpreted in pixel units. This search algorithm is recommended when the objects being sought are circular-type shapes, although the algorithm does have a tolerance for deformations of the circular shape. This controllable deformation tolerance (M_SAGITTA_TOLERANCE) could include detecting certain elliptical-type shapes as well as noisy circles. This search algorithm will typically find circular shapes faster and in a more robust manner than the search algorithms used for M_GEOMETRIC or M_GEOMETRIC_CONTROLLED types of Model Finder contexts. When using this type of Model Finder context, M_SEARCH_ANGLE_RANGE, M_SEARCH_POSITION_RANGE, and M_SEARCH_SCALE_RANGE are turned off by default and cannot be set to M_ENABLE. When using this type of context, you must also allocate an M_SHAPE_CIRCLE Model Finder result buffer.															

		c		e	f	g	a		i		j	k	l	m	n	b
M_SHAPE_ELLIPSE	Specifies that the Model Finder context uses an elliptical model search algorithm. This context only supports M_ELLIPSE type models, and only supports one model per context. The model cannot be calibrated. If the target is associated with a camera calibration context, the model will be interpreted in world units; conversely, if the target does not have a camera calibration context associated with it, the model will be interpreted in pixel units. This search algorithm is recommended when the objects being sought are elliptical-type shapes (it can also find circles). Besides controllable deformation tolerance (M_SAGITTA_TOLERANCE), it allows finding ellipses at varying aspect ratios (M_MODEL_ASPECT_RATIO and M_MODEL_ASPECT_RATIO_..._FACTOR). When using this type of Model Finder context, M_SEARCH_ANGLE_RANGE, M_SEARCH_POSITION_RANGE, and M_SEARCH_SCALE_RANGE are turned off by default and cannot be set to M_ENABLE. When using this type of context, you must also allocate an M_SHAPE_ELLIPSE Model Finder result buffer.																

M_SHAPE_RECTANGLE	Specifies that the Model Finder context uses a rectangular model search algorithm. This context only supports M_RECTANGLE type models, and only supports one model per context. The model cannot be calibrated. If the target is associated with a camera calibration context, the model will be interpreted in world units; conversely, if the target does not have a camera calibration context associated with it, the model will be interpreted in pixel units. This search algorithm is recommended when the objects being sought are rectangular-type shapes (it can also find squares). Besides controllable deformation tolerance (M_DEVIATION_TOLERANCE), it allows finding rectangles at varying aspect ratios. However, at least a portion of each of their sides must be visible; that is, the model coverage on any side cannot be 0 (M_MIN_SIDE_COVERAGE). When using this type of context, you must also allocate an M_SHAPE_RECTANGLE Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_SHAPE_SEGMENT	Specifies that the Model Finder context uses a segment model search algorithm. This context only supports M_SEGMENT type models, and only supports one model per context. The model cannot be calibrated. If the target is associated with a camera calibration context, the model will be interpreted in world units; conversely, if the target does not have a camera calibration context associated with it, the model will be interpreted in pixel units. This search algorithm is recommended when the objects being sought are segment-type shapes. Besides controllable deformation tolerance (M_DEVIATION_TOLERANCE), it allows finding segments lying on contours. When using this type of context, you must also allocate an M_SHAPE_SEGMENT Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

ContextIdPtr

Specifies the address of the variable in which to write the Model Finder context identifier or specifies the data type that the function should use to return the Model Finder context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated Model Finder context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated Model Finder context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_MOD_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the Model Finder context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the circular Model Finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated circular Model Finder context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the controlled geometric Model Finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated controlled geometric Model Finder context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the elliptical Model Finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated elliptical Model Finder context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the geometric Model Finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated geometric Model Finder context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the Model Finder context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_MOD_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodAllocResult

Synopsis

Allocate a Model Finder result buffer.

Syntax

```
AIL_ID MmodAllocResult(
    AIL_ID SysId,          //in
    AIL_INT64 ControlFlag, //in
    AIL_ID *ModResultIdPtr //out
)
```

Description

This function allocates a result buffer, on the specified system, to store results obtained from an MmodFind() operation.

When the Model Finder result buffer is no longer required, release it using MmodFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the ModelFinder result buffer's lifetime and you must not manually free it.

Parameters

SysId INQ

Specifies the system on which to allocate the result buffer.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag INQ

Specifies the type of result buffer to allocate. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of Model Finder result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies that the Model Finder result buffer can hold results from a search with an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of context.	c		e	f	g	a		i		j	k	l	m	n	b
M_SHAPE_CIRCLE	Specifies that the Model Finder result buffer can hold results from a search with an M_SHAPE_CIRCLE type of context.	c		e	f	g	a		i		j	k	l	m	n	b
M_SHAPE_ELLIPSE	Specifies that the Model Finder result buffer can hold results from a search with an M_SHAPE_ELLIPSE type of context.	c		e	f	g	a		i		j	k	l	m	n	b
M_SHAPE_RECTANGLE	Specifies that the Model Finder result buffer can hold results from a search with an M_SHAPE_RECTANGLE type of context.	c		e	f	g	a		i		j	k	l	m	n	b
M_SHAPE_SEGMENT	Specifies that the Model Finder result buffer can hold results from a search with an M_SHAPE_SEGMENT type of context.	c		e	f	g	a		i		j	k	l	m	n	b

ModResultIdPtr

Specifies the address of the variable in which to write the Model Finder result buffer identifier or specifies the data type that the function should use to return the Model Finder result buffer identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated Model Finder context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated Model Finder context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_MOD_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the Model Finder context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the geometric search Model Finder result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated geometric search Model Finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the shape search Model Finder result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated shape search Model Finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the Model Finder result buffer identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_MOD_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodControl

Synopsis

Control a Model Finder context, individual model setting, or Model Finder result buffer setting.

Syntax

```
void MmodControl(
    AIL_ID ContextOrResultId, //out
    AIL_INT Index,            //in
    AIL_INT64 ControlType,    //in
    AIL_DOUBLE ControlValue   //in
)
```

Description

This function allows you to control a setting of a Model Finder context, one (or all) of the models contained therein, or a Model Finder result buffer. These settings control the execution of MmodFind() operations. Most of the control type settings can be inquired using MmodInquire().

The units in which you should provide values (for example, position and width), depend on the model and context that you are using. In general, provide values as follows:

- For a non-synthetic model, provide values in the pixel coordinate system, even if the target is calibrated. The only exception is when specifying angles (for example, M_REFERENCE_ANGLE); in these cases, you must specify the angles in the relative coordinate system if the target is calibrated.
- For a synthetic model in an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED context, provide values in the user-defined units for the model; their relationship to pixels is defined using **MmodControl**() with M_PIXEL_SCALE. If the target is not calibrated, M_PIXEL_SCALE is used for the match. If the target is calibrated, model units are used; note that these should be the same as the calibrated units. In a few cases, instead of providing values in model units, you must provide values in pixel units (for example, M_POSITION_...); these cases are clearly indicated.
- For a model in an M_SHAPE_... context, provide values in model units, which in this case are the same as the target units. If the target is calibrated, the model's settings will be interpreted in world units. If the target is not calibrated, the model's settings will be interpreted in pixel units.

For each model defined in your Model Finder context, you can restrict the region in which to find the reference position of the model (the search region) to a rectangular subsection of your target image; this is useful if you know that occurrences can only be found in one region of the target image. There are three methods to do so:

- You can manually define the search region using successive calls to **MmodControl**() with M_SEARCH_POSITION_RANGE set to M_ENABLE, M_POSITION_X, M_POSITION_Y, M_POSITION_DELTA_NEG_X, M_POSITION_DELTA_NEG_Y, M_POSITION_DELTA_POS_X, and M_POSITION_DELTA_POS_Y. For models in M_SHAPE_... type Model Finder contexts, the M_ANGLE_REGION control type is also available, which allows you to set your rectangular search region at a non-zero angle.
- You can call **MmodControl**() with M_SEARCH_POSITION_RANGE set to M_ENABLE, M_SEARCH_POSITION_FROM_GRAPHIC_LIST, and the identifier of a 2D graphics list. The 2D graphics list must contain one graphic of type rectangle, and have its input units set to pixels; Aurora

Imaging Library will use its position and dimensions to define the search region. Note that for models in M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type Model Finder contexts, the rectangular graphic must be set with no angle, but for models in M_SHAPE_... type contexts, it can be set at any angle.

- For models in M_SHAPE_... type Model Finder contexts, you can call MmodFind() directly with a target image buffer that is associated with a rectangular region of interest at any angle.

Changing certain control type settings might require preprocessing of the Model Finder context again, using MmodPreprocess(). To know if the Model Finder context needs to be preprocessed, call MmodInquire() with the M_PREPROCESSED inquire type.

You can automatically define an image-type model that is based on specified control settings using this function with M_AUTO_DEFINE. To have specified settings taken into account during model definition, you must allocate an empty image-type model using MmodDefine() with M_AUTO_DEFINE, set the appropriate control types using **MmodControl**(), and then call **MmodControl**() with M_AUTO_DEFINE. M_AUTO_DEFINE defines a unique model by searching the model source image for unique geometric features that match the settings specified using **MmodControl**(). When the most suitable geometric features are found, the function automatically defines a model from those features. If none are found, no model is allocated and an error is reported. It can take several seconds to find the best model (or more for large or small images). To be useful, the model source image should be a typical target image; otherwise, the selected area for the model might never be present in the target image. For more information on defining an image-type model automatically, see the Image-type models subsection of the Defining and adding models to your Model Finder context section of Chapter 8: Geometric Model Finder.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MmodControl()** internally calls the *AIL_DOUBLE* version of this function (MmodControlDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MmodControlInt64()).

Parameters

ContextOrResultId

Specifies the Model Finder context or Model Finder result buffer whose settings you want to modify. The Model Finder context or result buffer must have been previously allocated on the system using MmodAlloc() or MmodAllocResult(), respectively.

Index

Specifies that the Model Finder context, an individual model, or a Model Finder result buffer is controlled. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a context, model, or result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies the default. If a Model Finder context is specified, same as M_ALL. If a Model Finder result buffer is specified, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Applies the specified control setting to all models, if a Model Finder context is specified. All the models in the Model Finder context must support the specified control type, otherwise, an error is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Controls a general setting of a specified Model Finder context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Controls a general setting of a Model Finder result buffer when ContextOrResultId is a result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER_MODELS	Specifies the index of the individual model to control, if a Model Finder context is specified. User labels cannot be used as indices; to retrieve the index of a model from its user label, use MmodInquire() with M_INDEX_FROM_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b

ControlType

Specifies the setting to change.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

All values are set in pixels, regardless of whether the model is calibrated.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters.

- For the context and models
- For the context
- For one or all models

- For a result buffer

The following ControlType and corresponding ControlValue settings can be specified for both the Model Finder context (M_CONTEXT) and each individual model (or all models) in the Model Finder context:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the context and models																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_NUMBER		Sets the number of occurrences for which to search. For M_CONTEXT, this control type sets the maximum number of all model occurrences for which to search in the target. For a model, this control type sets the maximum number of occurrences of the specified model, for which to search in the target. Once the number of occurrences for the context, with scores and target scores greater than or equal to their certainty levels, has been reached, the search will stop, regardless of whether the actual number set for any individual model has been found. To further illustrate the relationship between the M_NUMBER setting for the context and that of the individual models, see the Expected number of occurrences subsection of the Customizing search settings section of Chapter 8: Geometric Model Finder. INQ															
	M_DEFAULT	Specifies the default value. For M_CONTEXT, the default is M_ALL. For a model, the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies to find all occurrences. For M_CONTEXT, M_ALL finds all the occurrences specified for each model in the context. For a model, M_ALL finds all the occurrences of the specified model. Note that this setting can increase the search time; always set M_NUMBER to a specific number whenever possible.	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0	Specifies the number of occurrences. For M_CONTEXT, this value specifies the number of occurrences, for all models within the context, to find in the target. For a model, this value specifies the maximum number of occurrences of the specified model to find in the target.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue settings can be specified for the Model Finder context (M_CONTEXT):

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the context																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_ACCURACY		Sets the accuracy with which to find occurrences. The precision achieved is dependent on the quality of the model and the target. In addition, positional accuracy is slightly affected by the setting of the M_SPEED. Note that lower accuracy can speed up the MmodFind() operation. This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ															
	M_DEFAULT	Same as M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies high accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LOW	Specifies low accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEDIUM	Specifies medium accuracy.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTIVE_EDGELS		Sets the degree to which Aurora Imaging Library processes edgels in the target (MmodFind()) The target edgels that Aurora Imaging Library processes are considered active. By limiting the active edgels in the target, you can speed up the search. To draw the active edgels in the target, use MmodDraw() with M_DRAW_ACTIVE_EDGELS. This control type is only supported for a geometric Model Finder context (MmodAlloc() with M_GEOMETRIC). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 100.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the degree to which Aurora Imaging Library processes target edgels, as a percentage. Lower settings typically result in fewer active edgels in the target and faster searches, although accuracy can be compromised. Conversely, higher settings result in more active edgels in the target and higher accuracy, although searches can take longer.	c		e	f	g	a		i		j	k	l	m	n	b

M_ASPECT_RATIO		Sets the aspect ratio to apply to the target before starting the search. When the target is not calibrated, this control type allows you to quickly compensate for aspect ratio distortion in the target, typically a side effect of the sampling rate used by the frame grabber. When the target is calibrated, this control type is ignored. This control type is mainly useful when dealing with synthetic models since it does not compensate for aspect ratio distortion in the model. When performing the search, the nominal search aspect ratio is taken into account horizontally, adjusting the target pixel width to equal the pixel height. As such, results will be returned for this corrected target coordinate system. However, the results are converted to the original target coordinate system if you draw them using MmodDraw(). This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.5 <= *Value* <= 2.0	Specifies the pixel width/pixel height of the target.	c		e	f	g	a		i		j	k	l	m	n	b
M_DETAIL_LEVEL		Sets the level of details to extract from model images and target images during edge extraction. The detail level determines what is considered an edge/background. A higher detail level will include more edges than a lower detail level. This setting is used when dealing with image-type models and/or image-type targets. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b

	M_HIGH	Specifies a high level of detail.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEDIUM	Specifies a medium level of detail.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies a very high level of detail.	c		e	f	g	a		i		j	k	l	m	n	b
M_EDGE_ACCURACY		Controls the edge accuracy to be used when calculating the edges. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies a high level of edge accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies a very high level of edge accuracy.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIRST_LEVEL		Sets the resolution level for the initial stage (lowest resolution) of the search. Level 0 is the original target and each higher level is half the size (and resolution) of the previous one. If the specified level is not supported by the search algorithm, the highest possible level will be used. A higher first level speeds up the initial search but might make it less reliable because the model might not retain enough distinctive features at a lower resolution. M_FIRST_LEVEL is for advanced users of the Model Finder module. The default setting usually provides the best results for the search operation. Note that if you set a lower value for M_FIRST_LEVEL than the one assigned to M_LAST_LEVEL, **MmodControl**() will override your settings and automatically assign M_FIRST_LEVEL the same value as M_LAST_LEVEL. This implies that only one resolution level will be used for the search. This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies that the first resolution is determined automatically.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <= 7	Specifies the resolution level. Only integer values are accepted.	c		e	f	g	a		i		j	k	l	m	n	b

M_LAST_LEVEL		Sets the resolution level for the final stage (the highest resolution) of the search. Level 0 is the original target and each higher level is half the size (and resolution) of the previous one. If the specified level is not supported by the search algorithm, the highest possible level will be used. A higher last level speeds up the initial search but might make it less reliable and less accurate because the model might not retain enough distinctive features at a low resolution. M_LAST_LEVEL is for advanced users of the Model Finder module. The default setting usually provides the best results for the search operation. Note that if you set a higher value for M_LAST_LEVEL than the one assigned to M_FIRST_LEVEL, **MmodControl**() will override your settings and automatically assign M_FIRST_LEVEL the same value as M_LAST_LEVEL. This implies that only one resolution level will be used for the search. This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies that the last resolution is determined automatically.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <= 7	Specifies the resolution level. Only integer values are accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESOLUTION_COARSENESS_LEVEL		Sets the resolution of edge approximation. This control type is only valid for M_SHAPE_... Model finder contexts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 <= *Value* <= 100.0	Specifies the resolution. When set to 0.0, a very fine edge approximation is performed. When set to 100.0, a coarse edge approximation is performed.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_TARGET_EDGES		Sets whether to save the target edges in the result buffer. By enabling this control type, you can retrieve target chain information using MmodGetResult() and you can draw target edges using MmodDraw(). For an M_SHAPE_SEGMENT type of context, this control type must be enabled if you want to use the search results as a target for a subsequent search of a model defined in an M_SHAPE_RECTANGLE type of context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that target edges cannot be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that target edges can be saved.	c		e	f	g	a		i		j	k	l	m	n	b

M_SEARCH_ANGLE_RANGE	Sets whether to perform calculations specific to angular-range search strategies. Typically, to search for models within their specified angular range (M_ANGLE-M_ANGLE_DELTA_NEG to M_ANGLE + M_ANGLE_DELTA_POS) in the target, calculations specific to angular-range search strategies should be enabled for the context. These calculations are not required to search for models at their nominal angle (M_ANGLE). In addition, if you expect that the occurrences sought are close to their model's nominal angle, you can try disabling these calculations to see if Model Finder can still find the required occurrences. Disabling these calculations might speed up the search depending on the model and the target. Note that M_SEARCH_ANGLE_RANGE must be enabled to search for rotation-invariant non-synthetic models (for example, an image-type model of a circle). Also note that candidates can only be returned as occurrences if found within the angular range specified for their model. Therefore, whether you enable or disable the calculation, you can restrict which candidates are returned as occurrences by narrowing the angular range. This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. The default is the same as M_ENABLE if the Model Finder context uses a general geometric search algorithm. The default is the same as M_DISABLE if the Model Finder context uses a controlled geometric search algorithm.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that calculations specific to angular-range search strategies are disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that calculations specific to angular-range search strategies are enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_POSITION_RANGE		Sets whether to perform calculations specific to position-range search strategies. Typically, to search for models within their specified position range (M_POSITION_X, M_POSITION_Y, M_POSITION_DELTA_NEG_X, M_POSITION_DELTA_POS_X, M_POSITION_DELTA_NEG_Y, M_POSITION_DELTA_POS_Y) in the target, calculations specific to position-range search strategies should be enabled for the context. These calculations are not required to search for models at their nominal position (M_POSITION_X and M_POSITION_Y). In addition, if you expect that the occurrences sought are close to their model's nominal position, you can try disabling these calculations to see if Model Finder can still find the required occurrences. Disabling these calculations might speed up the search depending on the model and the target. Note that candidates can only be returned as occurrences if found within the position range specified for their model. Therefore, whether you enable or disable the calculations, you can restrict which candidates are returned as occurrences by narrowing the position range. Finally, if these calculations are disabled and no nominal position is specified (M_POSITION_X and M_POSITION_Y set to M_ALL), no occurrences will be found and no errors will be returned. This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that calculations specific to position-range search strategies are disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that calculations specific to position-range search strategies are enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_SCALE_RANGE		Sets whether to perform calculations specific to scale-range search strategies. Typically, to search for models within their specified scale range (M_SCALE x M_SCALE_MIN_FACTOR) to (M_SCALE x M_SCALE_MAX_FACTOR) in the target, calculations specific to scale-range search strategies should be enabled for the context. These calculations are not required to search for models at their nominal scale (M_SCALE). In addition, if you expect that the occurrences sought are close to their model's nominal scale, you can try disabling these calculations to see if Model Finder can still find the required occurrences. Disabling these calculations might speed up the search depending on the model and the target. Note that candidates can only be returned as occurrences if found within the scale range specified for their model. Therefore, whether you enable or disable the calculation, you can restrict which candidates are returned as occurrences by narrowing the scale range. This control type cannot be set to M_ENABLE if the Model Finder context is set to M_GEOMETRIC_CONTROLLED. This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that calculations specific to scale-range search strategies are disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that calculations specific to scale-range search strategies are enabled. Note that when these calculations are enabled, limiting the scale range to the minimum required generally decreases the search time.	c		e	f	g	a		i		j	k	l	m	n	b
M_SHARED_EDGES		Sets whether to enable sharing of edges between occurrences. This control type is not supported for a model defined in an M_SHAPE_CIRCLE, an M_SHAPE_ELLIPSE, or an M_SHAPE_SEGMENT type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the sharing of edges. When disabled, edges that are part of more than one occurrence are considered as part of the occurrence with the greatest score only.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to enable the sharing of edges.	c		e	f	g	a		i		j	k	l	m	n	b
M_SMOOTHNESS		Sets the degree of smoothness (noise reduction) applied to the model images and target images during the edge extraction. This setting is only used when dealing with image-type models or using image-type targets. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images. A setting of 0.0 indicates almost no noise reduction effect, while a setting of 100.0 indicates a very strong noise reduction effect.	c		e	f	g	a		i		j	k	l	m	n	b
M_SPEED		Sets the algorithm's search speed. Note that increasing the search speed can decrease the robustness and subpixel accuracy of the MmodFind() operation. This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies a high speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LOW	Specifies a low speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEDIUM	Specifies a medium speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies a very high speed.	c		e	f	g	a		i		j	k	l	m	n	b
M_STOP_FIND		Stops the current MmodFind() operation. You must call **MmodControl**() with M_STOP_FIND from another thread (typically of higher priority). The number of occurrences found at that point will be returned.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Implements the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b

M_TARGET_CACHING		Sets whether target caching is enabled. When target caching is enabled, the geometric representation of the target, generated by MmodFind(), is kept in the result. Subsequent calls to MmodFind() can reuse the representation, if they use the same result buffer and target. This increases the speed of the searches. Note that the geometric representation generated with a given Model Finder context might not be compatible with another context. In such a case, the cache will not be used. Also note that this constant cannot be set to M_ENABLE if the Model Finder context is set to M_GEOMETRIC_CONTROLLED. This control type is not supported for an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable target caching. Each call to MmodFind() will generate the geometric representation of the target.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to enable target caching in the result.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT		Sets the maximum search time for MmodFind(), in msec. If a search times out, the number of occurrences found at that point will be returned. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 2000.0 msec.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies an infinite amount of search time.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the maximum search time, in msecs.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue settings can be specified for each individual model or all models in the Model Finder context:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For one or all models																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_ACCEPTANCE		Sets the acceptance level for the score. An occurrence will be returned only if the match score between the target and the model is greater than or equal to this level. The score is a measure, as a percentage, of the presence and fit of the model's active edges in the occurrence. Using M_FIT_ERROR_WEIGHTING_FACTOR, you can specify how much the fit influences the score. INQ															
	M_DEFAULT	Specifies the default value; the default value is 60.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies an acceptable score, as a percentage. 100% indicates that for every active edge in the model, a corresponding edge must be found in the occurrence with a perfect fit (generally hard to obtain).	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_TARGET		Sets the acceptance level for the target score. An occurrence will be returned only if the target score between the target and the model is greater than or equal to this level. The target score is a measure, as a percentage, of edges found in the occurrence that are not found in the original model, weighted by the deviation in position of the common edges. Using M_FIT_ERROR_WEIGHTING_FACTOR, you can specify how much the fit influences the target score. This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0%.	c		e	f	g	a		i		j	k	l	m	n	b

0.0 <= *Value* <= 100.0	Specifies an acceptable target score, as a percentage. At 0.0%, any number of extra edges is tolerated (100% allows for no extra edges in the occurrence and the occurrence must have a perfect fit).	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE	Sets the nominal search angle; this is the angle at which you expect to find the model's reference axis (specified using M_REFERENCE_ANGLE) in the target. For a model defined in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context, the model's reference axis is, by default, aligned with its principle axis (its width). This control type is not supported for a model defined in an M_SHAPE_CIRCLE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 360.0	Specifies the nominal search angle, in degrees. If the Model Finder operation is performed on a calibrated target, you must specify the angle with respect to the relative coordinate system of the target.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_DELTA_NEG	Sets the lower limit of the angular range, relative to the nominal search angle (M_ANGLE). Occurrences with angles outside the angular-range cannot be returned as results. Note that typically, to search for model occurrences within the angular range, calculations specific to angular-range search strategies should be enabled (M_SEARCH_ANGLE_RANGE) for the context. When enabled, the angular range should be used to cover an expected variance in angle. Note that the actual angle of the occurrence does not affect search speed. If you need to search for a model at discrete angles only (for example, at intervals of 90 degrees), it is typically more efficient to define several models with different expected angles, than to search through the full angular range. This control type is not supported for a model defined in an M_SHAPE_CIRCLE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 180.0°.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 180.0	Specifies the lower limit of the angular range, in degrees. If the Model Finder operation is performed on a calibrated target, you must specify the angle in the relative coordinate system of the target.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_DELTA_POS		Sets the upper limit of the angular range, relative to the nominal search angle (M_ANGLE). Occurrences with angles outside the angular-range cannot be returned as results. Note that typically, to search for model occurrences within the angular range, calculations specific to angular-range search strategies should be enabled (M_SEARCH_ANGLE_RANGE) for the context. When enabled, the angular range should be used to cover an expected variance in angle. Note that the actual angle of the occurrence does not affect search speed. If you need to search for a model at discrete angles only (for example, at intervals of 90 degrees), it is typically more efficient to define several models with different expected angles, than to search through the full angular range. This control type is not supported for a model defined in an M_SHAPE_CIRCLE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 180.0°.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 180.0	Specifies the upper limit of the angular range, in degrees. If the Model Finder operation is performed on a calibrated target, you must specify the angle in the relative coordinate system of the target.	c		e	f	g	a		i		j	k	l	m	n	b

M_ANGLE_MULTIPLE_RANGE		Sets whether to search for occurrences in multiple angular ranges. This allows you to search for occurrences at their nominal angle and at an angle perpendicular to the nominal search angle. For a model in an M_SHAPE_SEGMENT type of context, this also allows you to search for occurrences with opposite segment gradient directions (for example, along an edge of a chessboard grid row). If you only want to search for occurrences at the nominal angle, M_ANGLE, these calculations are not required. This control type is only supported for a model in an M_SHAPE_RECTANGLE or M_SHAPE_SEGMENT type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_STEP_180.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that searching in multiple angular ranges is disabled. This value is only supported for a model in an M_SHAPE_SEGMENT type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STEP_90	Specifies to search for models at angular ranges at 90 degree steps from the nominal search angle. It specifies to search for models within the specified angular range (M_ANGLE-M_ANGLE_DELTA_NEG to M_ANGLE + M_ANGLE_DELTA_POS), and also within the angular range at all 90 degree rotations of the specified angular range (for example, M_ANGLE + 90 - M_ANGLE_DELTA_NEG to M_ANGLE + 90 + M_ANGLE_DELTA_POS) in the target.	c		e	f	g	a		i		j	k	l	m	n	b

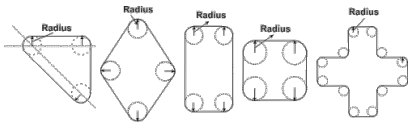
	M_STEP_180	Specifies to search for models at a 180 degree step. It specifies to search for models within the specified angular range (M_ANGLE-M_ANGLE_DELTA_NEG to M_ANGLE + M_ANGLE_DELTA_POS), and also within the angular range at a 180 degree rotation of the specified angular range (M_ANGLE + 180 - M_ANGLE_DELTA_NEG to M_ANGLE + 180 + M_ANGLE_DELTA_POS) in the target. Note that a rectangle occurrence found at an angle of M_ANGLE, and a rectangle occurrence found at an angle of M_ANGLE + 180 degrees, are considered the same rectangle.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_REGION		Sets the angle of the search region defined by the M_POSITION_... control types. Setting the angle of the search region does not set or affect the nominal search angle in any way; it only limits the location in which the reference position of the model can be found. If you expect to find occurrences at an angle, you must also set M_ANGLE. Likewise, searching within a region at an angle has no effect on the angle of a found occurrence's reference axis returned by MmodGetResult(). This control type is only valid for models in M_SHAPE_... contexts. For models in M_SHAPE_... contexts, you can alternatively set the search region at an angle by specifying M_SEARCH_POSITION_FROM_GRAPHIC_LIST and passing the identifier of a 2D graphics list containing a rotated rectangular graphic, or by passing a target image buffer associated with a rotated rectangular region of interest to MmodFind(). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0°.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angle of the search region, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_ASSOCIATED_CALIBRATION		Associates the specified camera calibration context with the specified model. Note that if you do not save, with the Model Finder context, the camera calibration contexts associated with the models (MmodSave() or MmodStream() with M_WITH_CALIBRATION), you must re-associate each model with its camera calibration context after restoring the context (MmodRestore(), or MmodStream()). In addition, for a synthetic model, you should use M_ASSOCIATED_CALIBRATION to associate the camera calibration context of a calibrated target with the model. Note that the camera calibration contexts associated with synthetic models are not saved or restored by MmodSave(), MmodRestore() or MmodStream(). This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_NULL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL	Removes the association between the model and a camera calibration context.	c		e	f	g	a		i		j	k	l	m	n	b
	Calibration context identifier	Specifies the camera calibration context to associate with the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO_DEFINE		Defines a unique model from a source image, automatically. This control type is only supported for models of type M_AUTO_DEFINE.	c		e	f	g	a		i		j	k	l	m	n	b
	Source image identifier	Specifies the identifier of the image buffer (model source image) from which to extract the model. The image buffer must be a 1-band 8-bit unsigned buffer.	c		e	f	g	a		i		j	k	l	m	n	b

M_BOX_MARGIN_BOTTOM	Sets a margin at the bottom of the bounding box of the model's active edges. The bounding box of the model's active edges plus the specified margins define the size of the model box. The model box is used to compute the target score and to set the different masks of the model. If you change this value, all of this model's previous masks are discarded, since their size is no longer valid. M_BOX_MARGIN_BOTTOM is only available for synthetic models. This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these contexts, the bottom margin is always 10.0% of the height of the bounding box of the model's active edges. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.0% of the height of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_MARGIN_LEFT	Sets a margin at the left side of the bounding box of the model's active edges. The bounding box of the model's active edges plus the specified margins define the size of the model box. The model box is used to compute the target score and to set the different masks of the model. If you change this value, all of this model's previous masks are discarded, since their size is no longer valid. M_BOX_MARGIN_LEFT is only available for synthetic models. This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these contexts, the left margin is always 10.0% of the width of the bounding box of the model's active edges. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.0% of the width of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b

	Value >= 0.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_MARGIN_RIGHT		Sets a margin at the right side of the bounding box of the model's active edges. The bounding box of the model's active edges plus the specified margins define the size of the model box. The model box is used to compute the target score and to set the different masks of the model. If you change this value, all of this model's previous masks are discarded, since their size is no longer valid. M_BOX_MARGIN_RIGHT is only available for synthetic models. This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these contexts, the right margin is always 10.0% of the width of the bounding box of the model's active edges. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0% of the width of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_MARGIN_TOP		Sets a margin at the top of the bounding box of the model's active edges. The bounding box of the model's active edges plus the specified margins define the size of the model box. The model box is used to compute the target score and to set the different masks of the model. If you change this value, all of this model's previous masks are discarded, since their size is no longer valid. M_BOX_MARGIN_TOP is only available for synthetic models. This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these contexts, the top margin is always 10.0% of the height of the bounding box of the model's active edges. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 10.0% of the height of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_CAD_Y_AXIS		Sets the direction of the Y-axis for a model of type M_DXF_FILE. When the model is added to the Model Finder context from a CAD DXF file, the model will retain its coordinate system (the origin and the axis). Most CAD DXF files are oriented so that the Y-axis is positive going up, whereas the coordinate system Aurora Imaging Library uses is oriented with the Y-axis positive going down. So if you take the edge coordinates of an object from a CAD DXF file and put them in an image, the imaged object will look flipped when compared to the original object. This is also the case for a model defined from a CAD DXF file. This means that, unless the object is symmetrical, the match will not be made. This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. The default value is M_FLIP for a model of type M_DXF_FILE. If the model is not of type M_DXF_FILE, this control type must remain set to M_DEFAULT, otherwise an error is generated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FLIP	Specifies to flip the Y-axis for the model so that the Y-axis is positive going down.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_FLIP	Specifies not to flip the Y-axis for the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY		Sets the certainty level for the score, as a percentage. If both the score and target scores are greater than or equal to their respective certainty levels, the occurrence is considered a match, without searching the rest of the target for better matches (provided the specified number of occurrences has been found). INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 90.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the certainty level for the score, as a percentage. If you set the certainty level too high (close to 100.0%), you slow down the search because you force the search algorithm to check the whole position range for the best possible match(es). A good certainty level is slightly lower than the expected score, so that the search can finish as soon as a match is found. However, if you set the certainty level too low, false matches might be found.	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY_TARGET		Sets the certainty level for the target score. If both the score and target scores are greater than or equal to their respective certainty levels, the occurrence is considered a match, without searching the rest of the target for better matches (provided the specified number of occurrences has been found). This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the certainty level for the target score, as a percentage. See the possible values of M_CERTAINTY for more details.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_RADIUS		Sets the radius used to round all the corners of predefined-shaped models that have corners.  This control type is only valid for models of type M_RECTANGLE, M_SQUARE, M_DIAMOND, M_TRIANGLE, and M_CROSS. Attempting to use this value for any other model type will generate an error. This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder result buffer. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the radius, in model units. If you set the radius to 0.0, there will be no rounding.	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX		Specifies the maximum expected model coverage. This control type is only supported for a model defined in an M_SHAPE_CIRCLE or M_SHAPE_ELLIPSE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage.	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MIN		Specifies the minimum expected model coverage. This control type is only supported for a model defined in an M_SHAPE_SEGMENT type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 40.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 100.0	Specifies the minimum expected model coverage.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEVIATION_TOLERANCE		Sets the tolerance for finding deformed rectangles or segments, for a model defined in an M_SHAPE_RECTANGLE or M_SHAPE_SEGMENT type of Model Finder context, respectively, given the other specified Model Finder constraints. This tolerance sets the allowable deformation of rectangles and line segments such that rectangles or segments that are not perfectly straight, or rectangles or segments that are broken up, would still be identified as potential occurrences. This control type is only supported for a model in an M_SHAPE_RECTANGLE or M_SHAPE_SEGMENT type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 25.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the tolerance as a percentage of the allowable deformation of a rectangle or segment, given the other Model Finder constraints. A value of 0.0% means that the rectangular shape or segment being sought needs to be as close as possible to a perfect model. A value of 100.0% means that the algorithm has the maximum tolerance for finding deformed rectangles or segments. This could lead to finding distorted rectangular shapes or segments.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_TOLERANCE_TO_CIRCLE		Sets the tolerance for approximating the aspect ratio of an ellipse occurrence to 1 (that is, reporting that the ellipse model matched a circle). Internally, the full precision of the aspect ratio of an occurrence is used to find the best occurrences of an ellipse. When you retrieve the resulting aspect ratio and height of the ellipse occurrence(s), using MmodGetResult() with M_ASPECT_RATIO and M_HEIGHT, the values are adjusted if the aspect ratio is close to 1 to account for pixelization and noise. If close enough to 1, the aspect ratio is returned as 1 and the height is returned as being the same as the width. You can use M_ELLIPSE_TOLERANCE_TO_CIRCLE to control how close to 1 the aspect ratio must be for it to be rounded to 1 and for the height to be adjusted. Aurora Imaging Library applies the tolerance to the aspect ratio between the ellipse's major and minor axis. M_ELLIPSE_TOLERANCE_TO_CIRCLE is only supported for a model defined in an M_SHAPE_ELLIPSE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.001.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 0.5	Specifies the tolerance applied to the ellipse occurrence's aspect ratio. If M_ASPECT_RATIO-1 is within the range (0, *Value*), then the ellipse is approximated as a circle.	c		e	f	g	a		i		j	k	l	m	n	b

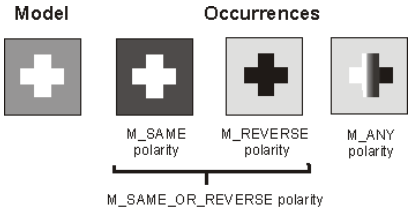
M_FIT_ERROR_WEIGHTING_FACTOR	Sets the fit error weighting factor. This factor determines the contribution of the fit error in the score and target score calculation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 25.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the fit error weighting factor, as a percentage. The higher the percentage, the greater the contribution of the fit error in determining the score and target score.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_SCORE_MIN	Sets the minimum expected occurrence fit score. This control type is only supported for a model defined in an M_SHAPE_... type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the minimum expected occurrence fit score.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_ANGLE	Sets the minimum angular separation required for two occurrences to be considered two distinct matches (two separate occurrences). Note, only one of the separation criteria needs to be met for occurrences to be considered distinct. DISTINCT OCCURRENCE = (Separation in X) OR (Separation in Y) OR (Separation in Angle) OR (Separation in Scale) OR (Separation in Aspect Ratio). This control type is not supported for a model defined in an M_SHAPE_CIRCLE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.0 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Disables the minimum angle separation criteria. When disabled (M_DISABLE), the angle is not a factor when determining if occurrences are distinct.	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 < *Value* <= 180.0	Specifies the minimum angular separation required, in degrees. This value is specified as an absolute angle value. If the Model Finder operation is performed on a calibrated target, you must specify the angle in the relative coordinate system of the target.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_ASPECT_RATIO		Specifies the minimum separation required in aspect ratios, for two occurrences to be considered distinct matches. This value is specified as an aspect ratio factor. Note, only one of the separation criteria needs to be met for occurrences to be considered distinct. DISTINCT OCCURRENCE = (Separation in X) OR (Separation in Y) OR (Separation in Angle) OR (Separation in Scale) OR (Separation in Aspect Ratio). This control type is only supported for a model in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the minimum aspect ratio separation criteria. When disabled, the aspect ratio is not a factor when determining if occurrences are distinct.	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 < *Value* <= 4.0	Specifies the criteria for minimum separation of aspect ratios.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_SCALE		Sets the minimum separation required in scale for two occurrences to be considered distinct matches (two separate occurrences). This value is specified as a scale factor. Note, only one of the separation criteria needs to be met for occurrences to be considered distinct. DISTINCT OCCURRENCE = (Separation in X) OR (Separation in Y) OR (Separation in Angle) OR (Separation in Scale) OR (Separation in Aspect Ratio). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.1.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies to disable the minimum scale separation criteria. When disabled, the scale is not a factor when determining if occurrences are distinct.	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 < *Value* <= 4.0	Specifies the criteria for minimum separation in scale.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_X		Sets the minimum separation required along the X-axis for two occurrences to be considered distinct matches (two separate occurrences). Note, only one of the separation criteria needs to be met for occurrences to be considered distinct. DISTINCT OCCURRENCE = (Separation in X) OR (Separation in Y) OR (Separation in Angle) OR (Separation in Scale) OR (Separation in Aspect Ratio). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the minimum separation in X criteria. When disabled, separation along the X-axis is not a factor when determining if occurrences are distinct.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the minimum separation as a percentage of the model's width at M_SCALE. This value can be greater than 100%.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_Y		Sets the minimum separation required along the Y-axis for two occurrences to be considered distinct matches (two separate occurrences). Note, only one of the separation criteria needs to be met for occurrences to be considered distinct. DISTINCT OCCURRENCE = (Separation in X) OR (Separation in Y) OR (Separation in Angle) OR (Separation in Scale) OR (Separation in Aspect Ratio). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0%.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies to disable the minimum separation in Y criteria. When disabled, separation along the Y-axis is not a factor when determining if occurrences are distinct.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the minimum separation as a percentage of the model's height at M_SCALE. This value can be greater than 100%.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SIDE_COVERAGE		Sets the minimum coverage on each of the sides of a rectangular model. Note that at least a portion of each side must be visible. This control type is only supported for a model in an M_SHAPE_RECTANGLE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 40.0.	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 <= *Value* <= 100.0	Specifies the minimum side coverage.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_ASPECT_RATIO		Sets the nominal aspect ratio factor for the model. The nominal aspect ratio factor is inversely applied to the height of the model, leaving the model's width constant. This control type is only supported for a model in an M_SHAPE_ELLIPSE or an M_SHAPE_RECTANGLE type of Model Finder context. For other types of Model Finder contexts, see M_ASPECT_RATIO. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_ASPECT_RATIO	Specifies the minimum possible value for this control type.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the value of the nominal aspect ratio for the model. > Note that the final aspect ratio must be greater than or equal to 1. That is to say that the width of the model divided by the height of the model, multiplied by the value set for this control type must be greater than or equal to 1.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODEL_ASPECT_RATIO_MAX_FACTOR		Sets the factor used to determine the upper limit (maximum permitted aspect ratio) of the model's aspect ratio. M_MODEL_ASPECT_RATIO is multiplied by this factor. Occurrences with aspect ratios outside of this range cannot be returned as results. This control type is only supported for a model in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.2.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies that the maximum factor for the aspect ratio of a model is infinite.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1.0	Specifies the factor that determines the maximum aspect ratio for a model.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_ASPECT_RATIO_MIN_FACTOR		Sets the factor used to determine the lower limit (minimum permitted aspect ratio) of the model's aspect ratio. M_MODEL_ASPECT_RATIO is multiplied by this factor. Occurrences with aspect ratios outside of this range cannot be returned as results. This control type is only supported for a model in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.8.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_ASPECT_RATIO	Specifies the minimum factor for the aspect ratio of a model.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 1.0	Specifies the factor that determines the minimum aspect ratio for a model.	c		e	f	g	a		i		j	k	l	m	n	b

M_PIXEL_SCALE	Sets the pixel scale of the model, if it is a synthetic model. This value is the scale to apply to the model to go from model units to pixel units. M_PIXEL_SCALE is used for the match if the target is not calibrated; if the target is calibrated, this value is not used for the match nor for the returned results. Regardless if calibrated, this value will affect MmodInquire() with M_ALLOC_SIZE_X and M_ALLOC_SIZE_Y, and it will also be used for draw operations. This is only valid for synthetic models. This control type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these types of contexts, model units are interpreted to be in the units of the target. If the target is calibrated, the units are interpreted to be in calibrated units. If the target is not, the units are interpreted to be in pixel units. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value of the pixel scale. If it is a calibrated model, the default is 1/(pixel size in X of the associated camera calibration). If it is not a calibrated model, the default is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the pixel scale. Note that if the model is not a synthetic model, the only valid value is 1.0. Any other value will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
M_POLARITY	Sets the expected polarity of occurrences, compared to that of the model.  This control type is not supported for a model defined in an M_SHAPE_SEGMENT type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value. For a model defined in an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, the default value is M_SAME. For a model defined in an M_SHAPE_CIRCLE, M_SHAPE_ELLIPSE, or M_SHAPE_RECTANGLE type of Model Finder context, the default value is M_SAME_OR_REVERSE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the occurrences can be a mixture of polarities.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REVERSE	Specifies that the polarity of occurrences is the reverse of that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the polarity of occurrences is the same as that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME_OR_REVERSE	Specifies that the polarity of occurrences can be either the same or the reverse of that of the model.	c		e	f	g	a		i		j	k	l	m	n	b

M_POSITION_DELTA_NEG_X		Sets the valid position range in the negative X-direction, relative to the nominal position (M_POSITION_X and M_POSITION_Y). The position range limits the region in which the position of a model occurrence can be found; position coordinates which fall outside this region cannot be returned as results (MmodGetResult() with M_POSITION_X and M_POSITION_Y). Note that the position returned for an occurrence is determined by the model's reference axis position. The region defined by the position range can lie partially, or totally, outside the target. Note that typically, to search for model occurrences within the position range, calculations specific to position-range search strategies should be enabled (M_SEARCH_POSITION_RANGE) for the context. When enabled, using a small position range generally decreases the search time, depending on the number of details present in the target. Always set the position range to the minimum required when speed is a consideration. When M_POSITION_X is set to M_ALL, M_POSITION_DELTA_NEG_X will be treated as if set to M_INFINITE regardless of its actual setting. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies the position range as the entire image plane in the negative X-direction. This means that any X-coordinate in the negative X-direction from the nominal position (even outside the target) can be returned as a result.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the position range's negative X-offset, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the offset in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b

M_POSITION_DELTA_NEG_Y		Sets the valid position range in the negative Y-direction, relative to the nominal position (M_POSITION_X and M_POSITION_Y). The position range limits the region in which the position of a model occurrence can be found; position coordinates which fall outside this region cannot be returned as results (MmodGetResult() with M_POSITION_X and M_POSITION_Y). Note that the position returned for an occurrence is determined by the model's reference axis position. The region defined by the position range can lie partially, or totally, outside the target. Note that typically, to search for model occurrences within the position range, calculations specific to position-range search strategies should be enabled (M_SEARCH_POSITION_RANGE) for the context. When enabled, using a small position range generally decreases the search time, depending on the number of details present in the target. Always set the position range to the minimum required when speed is a consideration. When M_POSITION_Y is set to M_ALL, M_POSITION_DELTA_NEG_Y will be treated as if set to M_INFINITE regardless of its actual setting. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies the position range as the entire image plane in the negative Y-direction. This means that any Y-coordinate in the negative Y-direction from the nominal position (even outside the target) can be returned as a result.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the position range's negative Y-offset, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the offset in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b

M_POSITION_DELTA_POS_X		Sets the valid position range in the positive X-direction, relative to the nominal position (M_POSITION_X and M_POSITION_Y). The position range limits the region in which the position of a model occurrence can be found; position coordinates which fall outside this region cannot be returned as results (MmodGetResult() with M_POSITION_X and M_POSITION_Y). Note that the position returned for an occurrence is determined by the model's reference axis position. The region defined by the position range can lie partially, or totally, outside the target. Note that typically, to search for model occurrences within the position range, calculations specific to position-range search strategies should be enabled (M_SEARCH_POSITION_RANGE) for the context. When enabled, using a small position range generally decreases the search time, depending on the number of details present in the target. Always set the position range to the minimum required when speed is a consideration. When M_POSITION_X is set to M_ALL, M_POSITION_DELTA_POS_X will be treated as if set to M_INFINITE regardless of its actual setting. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies the position range as the entire image plane in the positive X-direction. This means that any X-coordinate in the positive X-direction from the nominal position (even outside the target) can be returned as a result.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the position range's positive X-offset, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the offset in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b

M_POSITION_DELTA_POS_Y	Sets the valid position range in the positive Y-direction, relative to the nominal position (M_POSITION_X and M_POSITION_Y). The position range limits the region in which the position of a model occurrence can be found; position coordinates which fall outside this region cannot be returned as results (MmodGetResult() with M_POSITION_X and M_POSITION_Y). Note that the position returned for an occurrence is determined by the model's reference axis position. The region defined by the position range can lie partially, or totally, outside the target. Note that typically, to search for model occurrences within the position range, calculations specific to position-range search strategies should be enabled (M_SEARCH_POSITION_RANGE) for the context. When enabled, using a small position range generally decreases the search time, depending on the number of details present in the target. Always set the position range to the minimum required when speed is a consideration. When M_POSITION_Y is set to M_ALL, M_POSITION_DELTA_POS_Y will be treated as if set to M_INFINITE regardless of its actual setting. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies the position range as the entire image plane in the positive Y-direction. This means that any Y-coordinate in the positive Y-axis from the nominal position (even outside the target) can be returned as a result.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the position range's positive Y-offset, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the offset in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X	Sets the nominal search position for the X-coordinate. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ALL	Specifies all X-coordinates.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the nominal search position, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the position in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y		Sets the nominal search position for the Y-coordinate. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies all Y-coordinates.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the nominal search position, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the position in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_ANGLE		Sets the angle of the reference axis for the model. Angle results are returned relative to this axis, rather than to the model source image axis. For example, you can define a model from a source image and then search the same image by passing the image buffer to MmodFind(). If you then call MmodGetResult() with M_ANGLE to retrieve the angle of the occurrence, which is identical in shape and orientation to the model, the returned angle of the occurrence will match M_REFERENCE_ANGLE. By default, for an ellipse model defined in an M_SHAPE_ELLIPSE type of Model Finder context, the reference axis is aligned with the principal axis of the model (the width). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angle, in degrees. If the Model Finder operation is performed on a calibrated target, you must specify the angle with respect to the relative coordinate system of the target.	c		e	f	g	a		i		j	k	l	m	n	b

M_REFERENCE_X	Sets the X-offset of the origin of the model's reference axis, relative to the model origin. Position results return the X-coordinate of the model's reference axis origin transformed at the model occurrence. Position results are relative to the target origin. Note that the reference axis origin need not be in the model. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies that the default X-offset will be used. For a synthetic model, the default value is 0.0. For other types of models, the default value is the center of the model.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-offset value. The value is in pixels, and can be specified with subpixel accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_Y	Sets the Y-offset of the origin of the model's reference axis, relative to the model origin. Position results return the Y-coordinate of the model's reference axis origin transformed at the model occurrence. Position results are relative to the target origin. Note that the reference axis origin need not be in the model. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies that the default Y-offset will be used. For a synthetic model, the default value is 0.0. For other types of models, the default value is the center of the model.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-offset value. The value is in pixels, and can be specified with subpixel accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAGITTA_TOLERANCE	Sets the tolerance for finding deformed circles (allowable radii variation) for a model defined in an M_SHAPE_... type of Model Finder context, given the other specified Model Finder constraints. This control type is only supported for a model in an M_SHAPE_CIRCLE or M_SHAPE_ELLIPSE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 25.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the tolerance as a percentage of the allowable radii variation given the other Model Finder constraints. A value of 0.0% means that the circular shape being sought needs to be as close as possible to a perfect circle. A value of 100.0% means that the algorithm has the maximum tolerance for finding deformed circles. This could lead to finding elliptical shapes or circular type shapes with noise around the edges.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCALE		Sets the nominal search scale. For a model defined in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context, scale calculations are taken with respect to the model's width. The model's height is not taken into account. Use M_MODEL_ASPECT_RATIO to account for height differences. For example, if the model has a width of 4 and a height of 2 and the occurrence is expected to have a height of 2 and a width of 8, set M_SCALE to 2 and M_MODEL_ASPECT_RATIO to 2. Recall that the width of a model in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context, or the width of an occurrence of this type of model, is always measured along its principal axis. For a model defined in an M_SHAPE_SEGMENT type of Model Finder context, scale calculations are taken with respect to the model's length. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the value of the nominal search scale. For a synthetic model, the nominal scale can be any positive value. For all other models, the nominal scale must be from 0.5 to 2.0.	c		e	f	g	a		i		j	k	l	m	n	b

M_SCALE_MAX_FACTOR		Sets the factor used to determine the upper limit (maximum permitted scale) of the scale range. M_SCALE is multiplied by this factor. Occurrences with scales outside the scale-range cannot be returned as results. Note that typically, to search for model occurrences within the scale range, calculations specific to scale-range search strategies should be enabled (M_SEARCH_SCALE_RANGE) for the context. When enabled, the scale range should only cover the expected variance in scale to avoid slowing down the search and avoid finding unwanted occurrences. A search through a range of scales is performed in parallel, meaning that the actual scale of an occurrence has no bearing on which occurrence will be found first. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 2.0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies that the upper limit of the scale range is infinite. > This control value is only supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 <= *Value* <= 2.0	Specifies the factor that determines the maximum scale of the occurrence for a model in an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1.0	Specifies the factor that determines the maximum scale of the occurrence for a model in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b

M_SCALE_MIN_FACTOR	Sets the factor used to determine the lower limit (minimum permitted scale) of the scale range. M_SCALE is multiplied by this factor. Occurrences with scales outside the scale-range cannot be returned as results. Note that typically, to search for model occurrences within the scale range, calculations specific to scale-range search strategies should be enabled (M_SEARCH_SCALE_RANGE) for the context. When enabled, the scale range should only cover the expected variance in scale to avoid slowing down the search and avoid finding unwanted occurrences. A search through a range of scales is performed in parallel, meaning that the actual scale of an occurrence has no bearing on which occurrence will be found first. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.5.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 1.0	Specifies the factor that determines the minimum scale of the occurrence for a model in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
0.5 <= *Value* <= 1.0	Specifies the factor that determines the minimum scale of the occurrence for a model in an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_ASPECT_RATIO_CONSTRAINT	Sets whether to constrain candidates to the nearest bound of the aspect ratio range if the candidate is outside of the range. For example, if the maximum effective aspect ratio being sought is 2.0, and a candidate's aspect ratio is 2.3, and M_SEARCH_ASPECT_RATIO_CONSTRAINT is set to M_ENABLE, the candidate can be identified as an occurrence, with a reduced score, while if M_SEARCH_ASPECT_RATIO_CONSTRAINT is set to M_DISABLE, it is not. This control type is only supported for a model in an M_SHAPE_ELLIPSE type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that for a candidate to be considered an occurrence, it must have an aspect ratio within the defined aspect ratio range. Using this control value will limit the number of occurrences returned, but they will all conform to the defined aspect ratio range.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to consider candidates with an aspect ratio that falls outside of the specified aspect ratio range, but constrain their fit to the closest bound of the aspect ratio range, and reduce their score accordingly. This allows for more potential occurrences to be returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_POSITION_FROM_GRAPHIC_LIST		Specifies to use a search region defined from a 2D graphics list. The 2D graphics list must contain a maximum of one graphic of type rectangle (MgraRect(), MgraRectAngle(), MgraInteractive() with M_GRAPHIC_TYPE_RECT, or MgraRectFill()) and have its input units set to pixels (MgraControlList() with M_INPUT_UNITS set to M_PIXEL). For models in M_GEOMETRIC or M_GEOMETRIC_CONTROLLED contexts, the rectangular graphic in your 2D graphics list must be defined with an angle of 0 degrees. For models in M_SHAPE_... contexts, the rectangular graphic in your 2D graphics list can have any angle. Using this value is effectively equivalent to making multiple calls to **MmodControl**() with M_POSITION_X, M_POSITION_Y, M_POSITION_DELTA_NEG_X, M_POSITION_DELTA_NEG_Y, M_POSITION_DELTA_POS_X, M_POSITION_DELTA_POS_Y, and M_ANGLE_REGION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	2D graphics list identifier	Specifies the identifier of the 2D graphics list containing the rectangular region that will be used as the model search region.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEGMENT_CONSISTENT_GRADIENT		Sets whether the gradients of the segment model occurrences need to be consistent. This control type is only supported for a model in an M_SHAPE_SEGMENT type of Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the segment model occurrences can have the same gradients or reverse gradients.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the gradients of the segment model occurrences must be consistent.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_LABEL		Sets a unique user-defined label for the specified model. This label can be used as a means of identifying your model, independently from its index, in the Model Finder context. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_NO_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_LABEL	Specifies that no user label is associated with the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the user label of the model. The value must be an integer value that is not associated as a label with any other model in the Model Finder context. Since the same label cannot be associated with multiple models, you cannot pass a user label if the Index parameter of this function is set to M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue settings can be specified for a Model Finder result buffer:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a result buffer																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_MOD_DEFINE_COMPATIBLE		Saves all the necessary information in the result buffer to be able to define a model from a result occurrence (MmodDefine() with M_MOD_RESULT). This control type is not supported for an M_SHAPE_... type of Model Finder result buffer. INQ															

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to save any information.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to save all the necessary information.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_OUTPUT_UNITS		Sets whether to return results in pixels or world units. This essentially sets the output coordinate system to use. The setting of this control type will only affect functions within this module which return positional results. This control type can be changed at any time to return results in the required output units. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ACCORDING_TO_CALIBRATION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCORDING_TO_CALIBRATION	Specifies that results are returned in world units if the result was calculated on an image associated with a camera calibration context; otherwise, specifies that results are returned in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL	Specifies that results are returned in pixel units, with respect to the pixel coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD	Specifies that results are returned in world units, with respect to the relative coordinate system. If world units are specified, calling MmodGetResult() generates an error if the result was not calculated on a calibrated image.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MmodControlInt64 (AIL_ID ContextOrResultId, AIL_INT Index, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

ContextOrResultId

See ContextOrResultId of the main function for a description.

Index

See Index of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void MmodControlDouble (AIL_ID ContextOrResultId, AIL_INT Index, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

ContextOrResultId

See ContextOrResultId of the main function for a description.

Index

See Index of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodDefine

Synopsis

Add a model to, or delete a model from, a Model Finder context.

Syntax

```
void MmodDefine(
    AIL_ID ContextId,    //out
    AIL_INT64 ModelType, //in
    AIL_DOUBLE Param1,   //in
    AIL_DOUBLE Param2,   //in
    AIL_DOUBLE Param3,   //in
    AIL_DOUBLE Param4,   //in
    AIL_DOUBLE Param5    //in
)
```

Description

This function allows you to add a model to, or delete a model from, a Model Finder context. To an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, you can add a model from an image, from a result buffer, from two other models, or you can add a synthetic model. Image-type, result-type, merge-type, and synthetic models can be mixed in the same Model Finder context. To an M_SHAPE_... type of Model Finder context, you can only add a single synthetic model of the same type (for example, to an M_SHAPE_CIRCLE type of context, you can only add a single synthetic M_CIRCLE model).

There are two types of image-type models: models for which you manually define their location in the model source image (M_IMAGE), and models that Aurora Imaging Library automatically defines from the model source image (M_AUTO_DEFINE). M_AUTO_DEFINE defines a unique model by searching the model source image for unique geometric features. **MmodDefine**() can define an automatic model based on default settings. To have specified settings taken into account during model definition, you must allocate an empty image-type model using M_AUTO_DEFINE, set the appropriate control types using MmodControl(), and then call MmodControl() with M_AUTO_DEFINE. It can take several seconds to find the best model (or more for large or small images). To be useful, the model source image should be a typical target image; otherwise, the selected area for the model might never be present in the target image. For more information on defining a model automatically, see the Image-type models subsection of the Defining and adding models to your Model Finder context section of Chapter 8: Geometric Model Finder.

There are two types of result-type models: models defined from an Edge Finder result buffer (Edge Finder-type models), and models defined from a Model Finder result buffer (Model Finder-type model).

Merge-type models are composed of the edges of two different models. When you define a merge-type model, the two models from which it is merged are not affected.

There are two types of synthetic models: models of a predefined shape (predefined-shape models) and models defined from a CAD DXF file (CAD-file models). Use MmodDefineFromFile() to define a CAD-file model.

By default, synthetic models are defined to have a 10% margin around the bounding box of their active edges. For a model in an M_SHAPE_... type of Model Finder context, this margin is used for drawing. For a model in an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED context, any extra edges found in this area, when finding an occurrence, will reduce the target score. Regardless of the context, you can change the

size of the margins with the MmodControl() M_BOX_MARGIN_... control types.

How you specify the dimensions of synthetic models depends on the type of context:

- For an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, specify the dimensions of synthetic models in user-defined units. Then, specify the ratio between model units and pixel units using MmodControl() with M_PIXEL_SCALE. If the target is not calibrated, then M_PIXEL_SCALE is used for the match. If the target is calibrated, the model units should be the same as the calibrated units. If the target is calibrated, M_PIXEL_SCALE will be used for draw operations, but not for the match nor the returned results. If you are searching for a synthetic model in a calibrated target, you must also associate the camera calibration context of the target with the model, using M_ASSOCIATED_CALIBRATION. Aurora Imaging Library needs the camera calibration context for internal purposes at preprocessing time. Aurora Imaging Library will not use the camera calibration context to compensate for incongruencies between the model and the target, nor will Aurora Imaging Library map the model's coordinate system to that of the target.

  > After you have specified your pixel scale and scale settings, you can verify if your synthetic models are valid, using MmodInquire() with M_VALID. Synthetic models can be invalid if their global size in X or Y (*size of the model box* * M_PIXEL_SCALE * M_SCALE) is less than 16 or greater than 1024.

- For an M_SHAPE_... type of Model Finder context, specify the dimensions of the synthetic models in the units of the target; M_PIXEL_SCALE and M_ASSOCIATED_CALIBRATION are not used.

If the image or the result buffer used to define the model is calibrated, then that camera calibration context is automatically associated with the model. Models can use different camera calibration contexts within the same Model Finder context, provided all camera calibrations map to the same world. All image-type or result-type models must be either calibrated or not.

When a call to MmodFind() is made, all models that are part of the specified context are searched for simultaneously in the target.

Note, when a model is added or deleted, the Model Finder context must be preprocessed again, using MmodPreprocess().

The search is performed according to the general search settings specified in the Model Finder context (M_CONTEXT), as well as the individual model search settings. Both the context and individual model search settings can be specified using MmodControl().

The model index starts at 0, and each subsequent model added to the Model Finder context is given a sequential index number, in the order that the model was added. If a model is deleted, all entries with higher indices are shifted down one.

For more information on defining models, see the Guidelines for choosing models section of Chapter 8: Geometric Model Finder.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MmodDefine()** internally calls the *AIL_DOUBLE* version of this function (MmodDefineDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MmodDefineInt64()).

Parameters

ContextId

Specifies the Model Finder context to which to add, or from which to delete, the model. The Model Finder context must have been previously allocated on the required system using MmodAlloc().

ModelType INQ

Specifies the type of model to define when adding models to the context, or specifies to delete a model from the Model Finder context.

See the Parameter associations section for possible values that can be specified.

Param1

Specifies an attribute of the model to add. Its definition is dependent on the model type chosen.

See the Parameter associations section for possible values that can be specified.

Param2

Specifies an attribute of the model to add. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param3

Specifies an attribute of the model to add. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param4

Specifies an attribute of the model to add. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param5

Specifies an attribute of the model to add. Its definition is dependent on the model type chosen.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Parameter associations

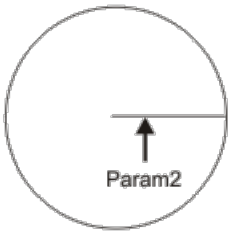
The tables below list possible values for the **ModelType**, **Param1**, **Param2**, **Param3**, **Param4**, and **Param5** parameters.

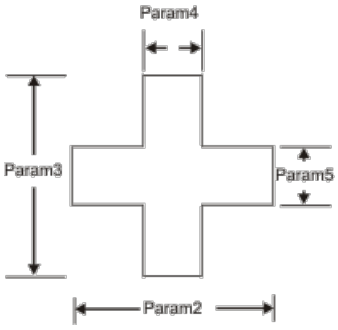
- For adding a model to the context
- For removing a model from the context

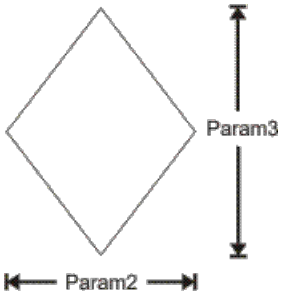
To add a model to the context, the ModelType, Param1, Param2, Param3, Param4, and Param5 parameters can be set to the following values. Note that any unused parameters should be set to **M_DEFAULT**.

For adding a model to the context

ModelType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Param1																
Param2																
Param3																
Param4																
Param5																
M_AUTO_DEFINE	Defines a unique model whose features are extracted from an automatically selected region of the model source image. To define a model automatically based on specified control settings, call this function with Param1 set to M_NULL. Then, specify control settings using MmodControl(). Once the controls are set, define the model using MmodControl() with M_AUTO_DEFINE. To define a model automatically based on default control settings, call this function with Param1 set to the identifier of the model source image. This type of model is not supported for M_SHAPE_... type of Model Finder contexts.															
Param1	Sets the identifier of the image buffer or specifies to define an empty model.	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies that an empty model should be defined.	c		e	f	g	a		i		j	k	l	m	n	b
Source image identifier	Sets the identifier of the image buffer (model source image) from which to extract a unique model. The image buffer must be a 1-band 8-bit unsigned buffer. This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b

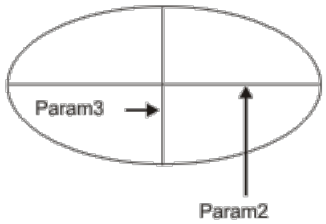
Param2	Sets the maximum displacement (shift), in the horizontal direction, expected between the position of the model in the source image and the position of the model when found in the target image. This information is used to select a model that is far enough from the image borders to be present in the target images.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Selects a model from anywhere in the whole width of the image.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the maximum expected pixel displacement in the horizontal direction.	c		e	f	g	a		i		j	k	l	m	n	b
Param3	Sets the maximum displacement (shift), in the vertical direction, expected between the position of the model in the source image and the position of the model when found in the target image. This information is used to select a model that is far enough from the image borders to be present in the target images.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Selects a model from anywhere in the whole height of the image.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the maximum expected pixel displacement in the verical direction.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE	Specifies a predefined circle as the model.  This type of model is only supported for an M_SHAPE_CIRCLE, M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Sets the foreground color of the circle. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies the default value. For an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, the default value is M_ANY. For an M_SHAPE_CIRCLE type of Model Finder context, the default value is M_FOREGROUND_BLACK.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ANY	Specifies that the model has no specific polarity. > This setting is not supported for an M_SHAPE_CIRCLE type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the radius of the circle. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_CROSS			Specifies a predefined cross as the model.  You can round the corners of this model using MmodControl() with M_CORNER_RADIUS. This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets the foreground color of the cross. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b

		M_ANY	Specifies that the model has no specific polarity.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the width of the cross. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	Param3		Sets the height of the cross. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	Param4		Sets the horizontal thickness of the cross. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	Param5		Sets the vertical thickness of the cross. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DIAMOND			Specifies a predefined diamond as the model.  You can round the corners of this model using MmodControl() with M_CORNER_RADIUS. This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets the foreground color of the diamond. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ANY	Specifies that the model has no specific polarity.	c		e	f	g	a		i		j	k	l	m	n	b

	M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
Param2		Sets the width of the diamond. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Param3		Sets the height of the diamond. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_EDGE_RESULT		Defines a model from the results of an edge extraction performed using Edge Finder. To define a model from an Edge Finder result buffer, the buffer must have been previously allocated using MedgeAllocResult(). MedgeCalculate() must have been already called. In addition, the Edge Finder result buffer must be compatible with the Model Finder context. Enable compatibility using MedgeControl() with M_MODEL_FINDER_COMPATIBLE prior to calling MedgeCalculate(). The model is defined to have all currently included edges in the result buffer. In the case of an Edge Finder-type model, the model source image is the image from which the Edge Finder results were obtained. This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
Param1		Sets the identifier of the Edge Finder result buffer. The Edge Finder result buffer (Param1) must be allocated on the same system as the Model Finder context (ContextId). If it is not, an error will occur.	c		e	f	g	a		i		j	k	l	m	n	b
Param2		Sets the X-offset of the model origin from the model source image.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies that the X-offset of the model origin is the origin of the model source image. The X-offset is 0.	c		e	f	g	a		i		j	k	l	m	n	b

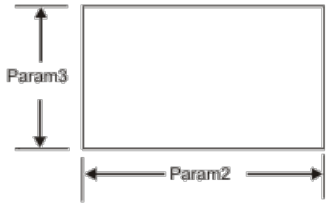
		Value	Specifies the X-offset of the model origin in the model source image, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Param3		Sets the Y-offset of the model origin from the model source image.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies that the Y-offset of the model origin is the origin of the model source image. The Y-offset is 0.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the Y-offset of the model origin in the model source image, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Param4		Sets the width of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies that the width of the model is the width of the model source image.	c		e	f	g	a		i		j	k	l	m	n	b
		16.0 <= *Value* <= 1024.0	Specifies the width of the model (in X), in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Param5		Sets the height of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies that the height of the model is the height of the model source image.	c		e	f	g	a		i		j	k	l	m	n	b
		16.0 <= *Value* <= 1024.0	Specifies the height of the model (in Y), in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

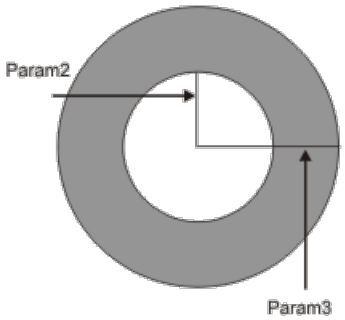
M_ELLIPSE		Specifies a predefined ellipse as the model.  Note that when specifying this type of model using an M_SHAPE_ELLIPSE type of Model Finder context, the width of the model must be larger than or equal to the height. This type of model is only supported for an M_SHAPE_ELLIPSE, M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Sets the foreground color of the ellipse. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. For an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, the default value is M_ANY. For an M_SHAPE_ELLIPSE type of Model Finder context, the default value is M_FOREGROUND_BLACK.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ANY	Specifies that the model has no specific polarity. > This setting is not supported for an M_SHAPE_ELLIPSE type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b

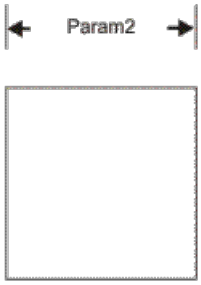
	Param2	Sets the width of the ellipse. For an M_SHAPE_ELLIPSE type of Model finder context, the width must be greater or equal to the height. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	Param3	Sets the height of the ellipse. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_IMAGE		Defines a model from the specified model source image. The default settings use the entire image as the model. For image-type models, the minimum and maximum model sizes are 16.0 x 16.0 and 4096.0 x 4096.0 pixels, respectively. To extract the model from a rotated region in the model source image, pass a model source image that is associated to a rectangular region of interest (ROI) at the required angle. You can associate an ROI to your source image using MbufSetRegion(). This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Sets the identifier of the image buffer (model source image) from which to extract the model. The image buffer must be a 1-band 8-bit unsigned buffer. This image buffer can have a rotated rectangular region of interest (ROI) associated with it, defined using MbufSetRegion() with a 2D graphics list, M_FILL_REGION, and either M_RASTERIZE or M_NO_RASTERIZE. An error will be generated if the ROI is only in raster format (M_RASTER) or is in a non-rectangular shape. When specifying an image buffer (model source image) that is associated with an ROI, you must set Param2, Param3, Param4, and Param5 to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2	Sets the X-offset of the model origin from the model source image. When specifying a model source image that is associated with an ROI, you must set this parameter to **M_DEFAULT**. INQ	c		e	f	g	a		i		j	k	l	m	n	b

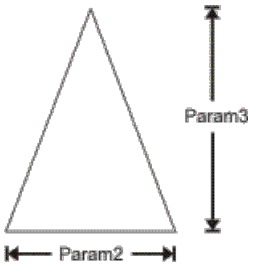
		M_DEFAULT	Specifies that the X-offset of the model origin is the origin of the model source image, that is, the X-offset is 0. Alternatively, if the model source image is associated with an ROI, it specifies to ignore this parameter.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the X-offset of the model origin in the source image, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Param3		Sets the Y-offset of the model origin. When specifying a model source image that is associated with an ROI, you must set this parameter to **M_DEFAULT**. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies that the Y-offset of the model origin is the origin of the model source image, that is, the Y-offset is 0. Alternatively, if the model source image is associated with an ROI, it specifies to ignore this parameter.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the Y-offset of the model origin in the source image, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Param4		Sets the width of the model. When specifying a model source image that is associated with an ROI, you must set this parameter to **M_DEFAULT**. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies that the width of the model is the distance from the origin of the model to the edge of the image in the X-direction. Alternatively, if the model source image is associated with an ROI, it specifies to ignore this parameter.	c		e	f	g	a		i		j	k	l	m	n	b
		16.0 <= *Value* <= 4096.0	Specifies the width of the model (in X), in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Param5		Sets the height of the model. When specifying a model source image that is associated with an ROI, you must set this parameter to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies that the height of the model is the distance from the origin of the model to the edge of the image in the Y-direction. Alternatively, if the model source image is associated with an ROI, it specifies to ignore this parameter.	c		e	f	g	a		i		j	k	l	m	n	b
		16.0 <= *Value* <= 4096.0	Specifies the height of the model (in Y), in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_MERGE_MODEL			Defines a model from two other models. These models have to be in the same Model Finder context. The model is defined with the edges of the first model and the edges of the second model that are not equal to the edges of the first model. The size of the merge-type model is the size of the first model. To include all of the edges of both models in the new model, set the first parameter to the index of the larger model. If the models being merged are calibrated, the new merge-type model will be associated with the camera calibration context of the first model. This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets the index of the first model. User labels cannot be used as indices; to retrieve the index of a model from its user label, use MmodInquire() with M_INDEX_FROM_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the index of the second model. User labels cannot be used as indices; to retrieve the index of a model from its user label, use MmodInquire() with M_INDEX_FROM_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b

M_MOD_RESULT	Defines a model from a Model Finder result buffer. The model is defined with the edges of the target at the position of the specified occurrence. To define a model from a Model Finder result buffer, enable MmodControl() with M_MOD_DEFINE_COMPATIBLE. In the case of a Model Finder-type model, the model source image is the target from which the results were obtained. This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Sets the identifier of the Model Finder result buffer. The Model Finder result buffer must be allocated on the same system as the Model Finder context (ContextId). If it is not, an error will occur.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Sets the index of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_RECTANGLE	Specifies a predefined rectangle as the model.  If adding the model to an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED context, you can round the corners of this model using MmodControl() with M_CORNER_RADIUS. This type of model is only supported for an M_SHAPE_RECTANGLE, M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Sets the foreground color of the rectangle. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies the default value. For an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, the default value is M_ANY. For an M_SHAPE_RECTANGLE type of Model Finder context, the default value is M_FOREGROUND_BLACK.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ANY	Specifies that the model has no specific polarity. > This setting is not supported for an M_SHAPE_RECTANGLE type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the width of the rectangle. For an M_SHAPE_RECTANGLE type of Model finder context, the width must be greater or equal to the height. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	Param3		Sets the height of the rectangle. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_RING			Specifies a predefined ring as the model.  This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets the foreground color of the ring. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ANY	Specifies that the model has no specific polarity.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the radius of the inner ring. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	Param3		Sets the radius of the outer ring. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_SEGMENT			Specifies a predefined segment as the model. This type of model is only supported for an M_SHAPE_SEGMENT type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets the length of the segment. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_SQUARE			Specifies a predefined square as the model.  You can round the corners of this model using MmodControl() with M_CORNER_RADIUS. This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets the foreground color of the square. INQ	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ANY	Specifies that the model has no specific polarity.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the width and height of the square. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_TRIANGLE			Specifies a predefined triangle as the model.  You can round the corners of this model using MmodControl() with M_CORNER_RADIUS. This type of model is not supported for M_SHAPE_... type of Model Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets the foreground color of the triangle. INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ANY	Specifies that the model has no specific polarity.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the width of the triangle. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	Param3	Sets the height of the triangle. INQ	c		e	f	g	a		i		j	k	l	m	n	b

The following ModelType parameter setting can be used to remove a model from the context.

Note that any unused parameters should be set to **M_DEFAULT**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For removing a model from the context																		
ModelType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	Param1																
	Param2																
	Param3																
	Param4																
	Param5																
M_DELETE		Deletes the specified model from the context.															
	Param1	Sets the model index. User labels cannot be used as indices; to retrieve the index of a model from its user label, use MmodInquire() with M_INDEX_FROM_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MmodDefineInt64 (AIL_ID ContextId, AIL_INT64 ModelType, AIL_INT64 Param1, AIL_DOUBLE Param2,
AIL_DOUBLE Param3, AIL_DOUBLE Param4, AIL_DOUBLE Param5)
```

Details

Parameters

ContextId

See ContextId of the main function for a description.

ModelType

See ModelType of the main function for a description.

Param1

See Param1 of the main function for a description.

Param2

See Param2 of the main function for a description.

Param3

See Param3 of the main function for a description.

Param4

See Param4 of the main function for a description.

Param5

See Param5 of the main function for a description.

```
void MmodDefineDouble (AIL_ID ContextId, AIL_INT64 ModelType, AIL_DOUBLE Param1, AIL_DOUBLE Param2,
AIL_DOUBLE Param3, AIL_DOUBLE Param4, AIL_DOUBLE Param5)
```

Details

Parameters

ContextId

See ContextId of the main function for a description.

ModelType

See ModelType of the main function for a description.

Param1

See Param1 of the main function for a description.

Param2

See Param2 of the main function for a description.

Param3

See Param3 of the main function for a description.

Param4

See Param4 of the main function for a description.

Param5

See Param5 of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodDefineFromFile

Synopsis

Defines a model from a file and adds it to a Model Finder context.

Syntax

```
void MmodDefineFromFile(
    AIL_ID ContextModId,         //out
    AIL_INT64 FileType,          //in
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_INT64 ControlFlag        //in
)
```

Description

This function defines a model from a file and adds it to a Model Finder context. Models defined from a file are CAD-type synthetic models; use MmodDefine() to add other types of models to the Model Finder context. This function does not support an M_SHAPE_... type of Model Finder context.

Note that the CAD DXF files must be in the R14 file format.

You must preprocess the Model Finder context after you add a model.

By default, synthetic models are defined to have a 10% margin around the bounding box of their active edges. When finding an occurrence, any extra edges found in this area will reduce the target score. You can change the size of the margins with the MmodControl() M_BOX_MARGIN_... control types.

When the model is added to the Model Finder context from a CAD DXF file, the model will retain the CAD-file's coordinate system (the origin and the axis). Specify the ratio between model units and pixel units using MmodControl() with M_PIXEL_SCALE.

> After you have specified your pixel scale and scale settings, you can verify if your synthetic models are valid, using MmodInquire() with M_VALID. Synthetic models can be invalid if their global size in X or Y (*size of the model box* * M_PIXEL_SCALE * M_SCALE) is less than 16 or greater than 1024.

Note that if the model is used to search in a calibrated target, you must also associate the camera calibration context of the target with the model, using M_ASSOCIATED_CALIBRATION. Aurora Imaging Library needs the camera calibration context for internal purposes at preprocessing time. Aurora Imaging Library will not use the camera calibration context to compensate for incongruencies between the model and the target, nor will Aurora Imaging Library map the model's coordinate system to that of the target.

If the target is calibrated, the model units should be the same as the calibrated units. If the target is calibrated, M_PIXEL_SCALE will be used for draw operations, but not for the match nor the returned results.

Most CAD DXF files are oriented so that the Y-axis is positive going up, whereas the coordinate system Aurora Imaging Library uses is oriented with the Y-axis positive going down. So if you take the edge coordinates of an object from a CAD DXF file and put them in an image, the imaged object will look flipped when compared to the original object. This is also the case for a model defined from a CAD DXF file. This means that, unless the object is symmetrical, the match will not be made. You can use M_CAD_Y_AXIS to set the Y-axis in the appropriate direction.

To delete a model from a Model Finder context, use MmodDefine() with M_DELETE.

Parameters

ContextModId

Specifies the identifier of the Model Finder context. For this parameter, you can only specify M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder contexts.

FileType

Specifies the type of file from which to define the model.

This parameter can be set to the following value:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of file																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DXF_FILE	Defines the model from the entities in the specified CAD DXF file. This function only supports version 14 2D CAD DXF files. This function does not support all entities that are possible in a CAD DXF file. If there are unsupported entities in the file, they are ignored, and the rest of the entities are read. The supported entities are as follows: LINE, POLYLINE, LWPOLYLINE, CIRCLE, ARC, ELLIPSE, INSERT, and BLOCK. Note that for the LWPOLYLINE entities, the bulge option is ignored by Aurora Imaging Library. A model defined from a CAD DXF file has no polarity.	c		e	f	g	a		i		j	k	l	m	n	b

FileName

Specifies the name and path of the file from which to define the model. The function handles (internally) the opening and closing of the file.

This parameter can be set to the following value:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
AIL_TEXT("*FileName*") [1]	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*", for example: "*remote:///C:\mydirectory\myfile*".	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

ControlFlag

Reserved for future expansion. Set this parameter to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodDraw

Synopsis

Draw specific features of models or result occurrences in an image buffer or 2D graphics list.

Syntax

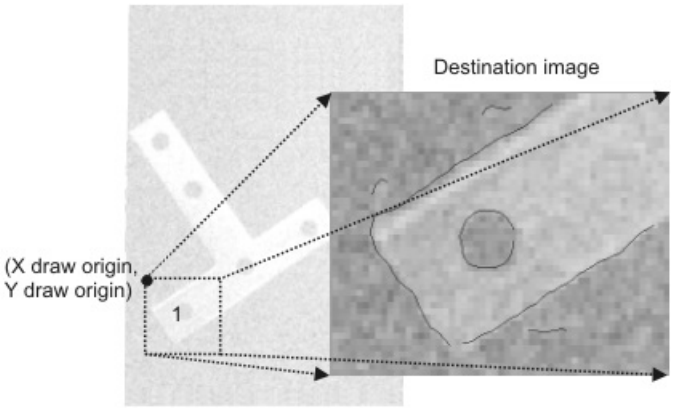
```
void MmodDraw(
    AIL_ID ContextGraId,           //in
    AIL_ID ContextOrResultModId,   //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_INT64 Operation,           //in
    AIL_INT Index,                 //in
    AIL_INT64 ControlFlag          //in
)
```

Description

This function draws specific features of models or result occurrences in the destination image buffer or 2D graphics list.

You can draw results and settings, obtained relative to an offset, at the top-left corner of the destination image, using MgraControl() with M_DRAW_OFFSET_X and M_DRAW_OFFSET_Y and zoom them using MgraControl() with M_DRAW_ZOOM_X and M_DRAW_ZOOM_Y. For more information, see the Drawing graphics with offset and zoom subsection of the Drawing graphics section of Chapter 26: Generating graphics.

Drawing a zoomed region of the model image



1 How much of this is drawn in the destination image is determined by the scale factor and the size of the destination image.

When zooming, **MmodDraw**() will draw into the destination image buffer even if the buffer is not large enough to contain all of the zoomed image. The image will be truncated.

Note that if you try to draw a model's edges and its Model Finder context has not been preprocessed, a preliminary edge extraction operation will be performed for the model.

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context to use when drawing. This parameter must be set to one of the following values:

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ContextOrResultModId

Specifies the Model Finder context or result buffer from which to extract the features to draw. The Model Finder context or result buffer must have been previously allocated using MmodAlloc() or MmodAllocResult(), respectively.

DstImageBufOrListGraId

Specifies the identifier of the destination image buffer or 2D graphics list in which to draw. The buffer can be any valid 1- or 3-band image buffer allocated using MbufAlloc...(). The 2D graphics list must be previously allocated using MgraAllocList(). By drawing into a display's overlay buffer or associating the 2D graphics list with the display, you can also annotate an image non-destructively.

If specifying an image buffer, it must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

When using **MmodDraw**() to draw features extracted from models or targets that are associated with a camera calibration context, the destination drawing buffer must also be calibrated to the same world coordinate system. **MmodDraw**() draws all the requested features according to whether the destination image buffer is physically corrected or not.

When drawing a model, the destination image must be at least the size of the model. Use MmodInquire() with the M_ALLOC_SIZE... inquire types to get the correct size.

When drawing a synthetic model, the destination image must have a size that matches the size of the model box scaled by M_PIXEL_SCALE.

Operation

Specifies the type of operation to perform. Operations can be added together to draw multiple features at a time. For example, to draw both the result occurrence's position and active edges, you would set the Operation parameter to M_DRAW_POSITION + M_DRAW_EDGES. The Operation parameter can be set to a combination of the following values:

The following operations can extract the features to draw from a model context only and cannot draw in a 2D graphics list.

Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying an operation that can extract features from a model context only																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_DONT_CARE +	Draws the model's "don't care" pixels. This operation is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FLAT_REGIONS +	Draws the model's "flat regions". This operation is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_IMAGE +	Draws the model image.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_WEIGHT_REGIONS +	Draws the model's weighted region mask. This operation is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b

The following operation can extract the features to draw from a result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying an operation that can extract features from a result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DRAW_ACTIVE_EDGELS +	Draws the active edgels of the target. To set the degree by which Aurora Imaging Library considers edgels in the target active, use MmodControl() with M_ACTIVE_EDGELS. When using M_DRAW_ACTIVE_EDGELS, you must set the Index parameter to M_GENERAL; also, the find operation must have been performed with M_SAVE_TARGET_EDGES set to M_ENABLE. For each model, Aurora Imaging Library establishes one set of active edgels in the target. For multi-model contexts, M_DRAW_ACTIVE_EDGELS draws all the edgels that have contributed to at least one model. You can only use M_DRAW_ACTIVE_EDGELS with a Model Finder result buffer whose results were generated using a geometric Model Finder context (MmodAlloc() with M_GEOMETRIC).	c		e	f	g	a		i		j	k	l	m	n	b

Combination values that can be used alone or as combination values for the values listed in For specifying an operation that can extract features from a model context only, For specifying an operation that can extract features from a result buffer.

You can use one or more of the following values in combination with each other or with the above-mentioned values, to specify an operation that can extract features to draw from both a model context and a result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying an operation that can extract features to draw from both a model context and a result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_BOX	Draws the model box, or a bounding box around the occurrence. Note that when drawing an occurrence, the bounding box is drawn maintaining the angle and scale of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_EDGES +	Draws the edges related to the model or result. When working with a model context, Aurora Imaging Library draws the active edges of the model (not masked out). When working with result occurrences, the drawing depends on whether you are adding M_MODEL or M_TARGET to M_DRAW_EDGES.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_POSITION	Draws a cross-like symbol at the model's reference axis origin or occurrence position. The cross is drawn maintaining the angle of the model's reference axis or the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_DRAW_EDGES.

You can add one of the following values to the above-mentioned value to specify whether to draw the active edges of the model transformed at the occurrence position or to draw the edges of the target in the region of the occurrence.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For M_DRAW_EDGES when working with result occurrences																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_MODEL	Draws the active edges of the model transformed at the occurrence position. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TARGET	Draws the edges of the target in the region of the occurrence. If the Index parameter is set to M_GENERAL, then all the edges of the target are drawn. To draw the edges of the target, the find operation must have been performed with M_SAVE_TARGET_EDGES set to M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b

Index

Specifies the index of the model in the specified Model Finder context, or of the result occurrence. User labels cannot be used as indices; to retrieve the index of a model from its user label, use MmodInquire() with M_INDEX_FROM_LABEL.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the model or of the result occurrence																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Refers to index 0 for models and M_ALL for results.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Draws the specified feature of all models or all occurrences.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Draws the specified feature of the entire target. M_GENERAL can only be specified when using M_DRAW_EDGES + M_TARGET or M_DRAW_ACTIVE_EDGELS.	c		e	f	g	a		i		j	k	l	m	n	b

Value >= 0	Specifies the model index or the result occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Specifies where in the destination image buffer to draw.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying where to draw																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Draws at the center of the top-left pixel of the destination image buffer; or, if drawing in a 2D graphics list, from the center of the top-left pixel of the image used at the time of annotation. When drawing features of a result occurrence, this parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_ORIGINAL	Draws at the offsets used to define the model region in the model source. If M_ORIGINAL is selected, the drawing is done at the original position of the model. M_ORIGINAL is only supported for image-type or Edge Finder-type models.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodFind

Synopsis

Search for the model(s) that are in the specified Model Finder context in a target image buffer, Edge Finder result buffer, or M_SHAPE_SEGMENT Model Finder result buffer.

Syntax

```
void MmodFind(
    AIL_ID ContextId,             //in
    AIL_ID TargetImageOrResultId, //in
    AIL_ID ModResultId            //out
)
```

Description

This function searches for the model(s) that are in the specified Model Finder context, in the specified target image, Edge Finder result buffer, or M_SHAPE_SEGMENT Model Finder result buffer; the latter case is only available when searching for the synthetic rectangular model of an M_SHAPE_RECTANGLE type of context. Results are stored in the specified Model Finder result buffer. The resulting values can be read using the MmodGetResult() function.

For each model defined in your Model Finder context, you can restrict the region in which to find the reference position of the model (the search region) to a rectangular subsection of your target image; this is useful if you know that occurrences can only be found in one region of the target image. There are three methods to do so:

- You can manually define the search region using successive calls to MmodControl() with M_SEARCH_POSITION_RANGE set to M_ENABLE, M_POSITION_X, M_POSITION_Y, M_POSITION_DELTA_NEG_X, M_POSITION_DELTA_NEG_Y, M_POSITION_DELTA_POS_X, and M_POSITION_DELTA_POS_Y. For models in M_SHAPE_... type Model Finder contexts, the M_ANGLE_REGION control type is also available, which allows you to set your rectangular search region at a non-zero angle.
- You can call MmodControl() with M_SEARCH_POSITION_RANGE set to M_ENABLE, M_SEARCH_POSITION_FROM_GRAPHIC_LIST, and the identifier of a 2D graphics list. The 2D graphics list must contain one graphic of type rectangle, and have its input units set to pixels; Aurora Imaging Library will use its position and dimensions to define the search region. Note that for models in M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type Model Finder contexts, the rectangular graphic must be set with no angle, but for models in M_SHAPE_... type contexts, it can be set at any angle.
- For models in M_SHAPE_... type Model Finder contexts, you can call **MmodFind**() directly with a target image buffer that is associated with a rectangular region of interest at any angle.

Setting the angle of the search region does not set or affect the nominal search angle in any way; if you expect to find occurrences at an angle, you must also set M_ANGLE. Likewise, searching within a region at an angle has no effect on the angle of a found occurrence's reference axis returned by MmodGetResult().

When searching in an Edge Finder result buffer, **MmodFind**() will search for the models only in the included edges of the Edge Finder result buffer. The search is performed using both the global search settings of the Model Finder context and each model's current search settings (see MmodControl()).

If an occurrence has both a score and target score greater than or equal to its model's respective certainty levels (specified using MmodControl() with M_CERTAINTY and M_CERTAINTY_TARGET), it is automatically considered an occurrence (default 80%); the remaining occurrences will be the best of those greater than or equal to the model's acceptance levels (specified using MmodControl() with M_ACCEPTANCE and M_ACCEPTANCE_TARGET).

The maximum number of occurrences found depends on the maximum number specified for the context (specified using MmodControl() with Index set to M_CONTEXT and ControlType set to M_NUMBER). If it is set to M_ALL, **MmodFind**() tries to find the maximum number of occurrences specified for each model (specified using MmodControl() with Index set to an index value and ControlType set to M_NUMBER), greater than or equal to the acceptance levels.

Note, the Model Finder context must be preprocessed (MmodPreprocess()) before calling this function.

If you are using an M_SHAPE_... type of Model Finder context, the model is considered to have the same units as the target. If the target is calibrated, the model will be interpreted in world units. Conversely, if the target does not have a camera calibration context associated with it, the model will be interpreted in pixel units. The model will be defined in the same coordinate system as the target. You cannot associate a calibration context with the model.

For other types of Model Finder contexts, their models can be calibrated. If their models are calibrated, you must specify a calibrated target image or Edge Finder results. Conversely, if the models are uncalibrated, the target must be uncalibrated as well. Note, however, that if you calibrate the models using McalUniform() to apply a uniform camera calibration of pixel size equal to 1, the target image or Edge Finder results need not be calibrated. Conversely, if the models are uncalibrated, they can be used with a calibrated target only if it has been calibrated using McalUniform() to apply a uniform camera calibration of pixel size equal to 1.

If the target image or Edge Finder results are calibrated, results are calculated in the world coordinate system; otherwise, they are calculated in the pixel coordinate system.

If the target image or Edge Finder results are calibrated, you can retrieve results in real-world or in pixel units. However, in the presence of distortion, some results are meaningless when converted from real-world to pixel units (for example, angle, scale, and transformation coefficient results). For example, if a model appears warped in the target image, but the camera calibration context of the target image compensates for this during the model search, the resulting angle is meaningful in the real-world coordinate system, and meaningless in the pixel coordinate system.

For an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, if the target image or Edge Finder results are not calibrated and the nominal search aspect ratio is not set to 1 (MmodControl() with M_ASPECT_RATIO), the nominal search aspect ratio is taken into account horizontally when performing the search, adjusting the target pixel width to equal the pixel height (the model's aspect ratio is not adjusted). As such, results will be returned for this corrected target coordinate system.

If you are searching for model(s) in an Edge Finder result buffer, you can speed up the search by setting M_EXTRACTION_SCALE in MedgeControl() prior to calling MedgeCalculate(). For more information on speeding up your search, see the Interfacing with Geometric Model Finder or Advanced Geometric Matcher section of Chapter 10: Edge Finder.

Parameters

ContextId

Specifies the Model Finder context to use for the search. The Model Finder context must have been

Specifies the Model Finder context to use for the search. The Model Finder context must have been previously allocated on the system using MmodAlloc().

TargetImageOrResultId

Specifies the target image, Edge Finder result buffer, or M_SHAPE_SEGMENT Model Finder result buffer in which to search for the model(s).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the target in which to search																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Edge Finder result buffer identifier	Specifies the identifier of the Aurora Imaging Library Edge Finder result buffer in which to search for the model(s). The Edge Finder result buffer must have been previously allocated on the required system using MedgeAllocResult(); MedgeCalculate() must have already been called. In addition, the Edge Finder result buffer must be compatible with the Model Finder context. Enable compatibility using MedgeControl() with M_MODEL_FINDER_COMPATIBLE prior to calling MedgeCalculate().	c		e	f	g	a		i		j	k	l	m	n	b
Image buffer identifier	Specifies the identifier of the Aurora Imaging Library image buffer in which to search for the model(s). The target image must be a 1-band 8-bit unsigned image. The minimum and maximum target image sizes are 16x16 and 32768x32768 pixels, respectively. It is important to note that the minimum and maximum target image sizes are Aurora Imaging Library limits. To make sure that you have enough memory, you can call **MmodFind**(); if you don't have enough memory, you will get an Aurora Imaging Library error. If your Model Finder context is of type M_SHAPE_CIRCLE, M_SHAPE_ELLIPSE, M_SHAPE_RECTANGLE, or M_SHAPE_SEGMENT, the target image buffer can have a rectangular region of interest (ROI) defined, with or without	c		e	f	g	a		i		j	k	l	m	n	b

	rotation. Otherwise, using an image buffer with an ROI will cause an error. The ROI defines the search region during the **MmodFind**() operation. This means that, while performing the search, **MmodFind**() ignores the M_POSITION_... and M_ANGLE_REGION control types. The region must be defined in vector format from a 2D graphics list (M_VECTOR or M_VECTOR_AND_RASTER). An error will be generated if the ROI is only in raster format (M_RASTER) or a non-rectangular shape. Searching within a rotated region does not affect the angle at which Aurora Imaging Library expects to find occurrences in any way; it only limits the location in which the reference position of the model can be found. If you expect to find occurrences at non-zero angles within a search region, you should specify the nominal search angle, using MmodControl() with M_ANGLE. Likewise, searching within a region at an angle has no effect on the angle of a found occurrence's reference axis returned by MmodGetResult().															
Segment-specific Model Finder result buffer identifier	Specifies the identifier of the M_SHAPE_SEGMENT Model Finder result buffer in which to search for a synthetic rectangular model; the model must be in an M_SHAPE_RECTANGLE type of context. The segment-specific result buffer must have been previously allocated on the required system using MmodAllocResult() with M_SHAPE_SEGMENT; **MmodFind**() must have already been called. In addition, target edges must have been stored in the result buffer (that is, MmodControl() with M_SAVE_TARGET_EDGES must have been enabled prior to calling **MmodFind**().	c		e	f	g	a		i		j	k	l	m	n	b

ModResultId

Specifies the result buffer in which to write the results of the search. If you are using an M_SHAPE_... type of Model Finder context, you need to specify a corresponding M_SHAPE_... type of Model Finder result buffer.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodFree

Synopsis

Free a Model Finder context or a result buffer.

Syntax

```
void MmodFree(
    AIL_ID ObjectId //in
)
```

Description

This function deletes the specified Model Finder context (and all its models) or result buffer identifier, and releases any memory associated with it. Note that to only delete individual models in the context, use MmodDefine().

All Model Finder contexts and all result buffers allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

Parameter

ObjectId

Specifies the identifier of the Model Finder context or result buffer to free. These must have been successfully allocated (with MmodAlloc() or MmodAllocResult()) prior to calling this function.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodGetResult

Synopsis

Get the specified type of result(s) from a Model Finder result buffer.

Syntax

```
void MmodGetResult(
    AIL_ID ResultId,      //in
    AIL_INT ResultIndex,  //in
    AIL_INT64 ResultType, //in
    void *ResultArrayPtr  //out
)
```

Description

This function retrieves the result(s) of the specified type from a Model Finder result buffer. Results are only available after calling MmodFind().

The result entries will be ordered by match score, starting with the highest score (regardless of the result type requested). When searching for multiple models, results are not sorted according to which model occurrence was found; results are still sorted according to score. The M_INDEX result type returns the index of the model associated with the Model Finder result; whereas the M_USER_LABEL result type returns its user label.

If your target image was associated with a camera calibration context, positional and dimensional results are, by default, returned with respect to the relative coordinate system of the image. Otherwise, these results are returned in pixels, relative to the top-left pixel in the target image.

> If your target image was associated with a camera calibration context but you want to retrieve positional and dimensional results in pixel units, use MmodControl() with the M_RESULT_OUTPUT_UNITS control type set to M_PIXEL. However, note that if you set M_RESULT_OUTPUT_UNITS to M_WORLD without specifying a calibrated image in which to calculate the results, **MmodGetResult**() will generate an error.

If the target is not calibrated and the nominal search aspect ratio is not set to 1 (MmodControl() with M_ASPECT_RATIO), results are returned for the corrected target coordinate system (whereby the target pixel width is equal to the pixel height).

Parameters

ResultId

Specifies the identifier of the Model Finder result buffer from which to retrieve results.

ResultIndex

Specifies where to get results. This parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying where to get results

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ALL for result types that apply to one or all occurrences. Same as M_GENERAL for general result types.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Retrieves the results of the specified type for all model occurrences found.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Retrieves a result relating to the entire Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER	Specifies the result index.	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result to retrieve.

See the Parameter associations section for possible values that can be specified.

ResultArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]
- array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]
- array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]
- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write results.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **ResultType** parameter and possible values returned to the **ResultArrayPtr** parameter.

- For a general result
- For an individual occurrence, all occurrences, or a general result
- For an individual occurrence or a general result
- For a result of a single occurrence or all occurrences
- For a result of a single occurrence or all occurrences (with constraints)
- For retrieving a transformation coefficient

When retrieving a general result, the ResultType parameter can be set to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a general result																	
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ResultArrayPtr - Possible values returned																
M_CONTEXT_ID +		Retrieves the identifier of the Model Finder context used by MmodFind() to obtain the results in the result buffer. Note that this identifier might not be valid if the Model Finder context has been freed using MmodFree().															
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_MOD_DEFINE_COMPATIBLE +		Retrieves whether the result can be used to define a model. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the result cannot be used to define a model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the result can be used to define a model.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER +		Retrieves the number of occurrences found (all models).	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT_END +		Retrieves whether the timeout has been reached. You can set the timeout limit using MmodControl() with M_TIMEOUT. By default, there is no limit.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the timeout has not been reached.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRUE	Specifies that the timeout has been reached.	c		e	f	g	a		i		j	k	l	m	n	b

When retrieving a general result, a result for an individual occurrence, or a result for all occurrences (M_ALL), the ResultType parameter can be set to the following value.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For an individual occurrence, all occurrences, or a general result																	
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ResultArrayPtr - Possible values returned																
M_NUMBER_OF_CHAINED_EDGELS +		Retrieves the total number of chained edgels. For a general result, M_NUMBER_OF_CHAINED_EDGELS returns the number of chained edgels in the target. For a specified occurrence, M_NUMBER_OF_CHAINED_EDGELS returns the chained edgels in the occurrence. When used with M_ALL, an array of results is returned, with one result per occurrence. You can use M_NUMBER_OF_CHAINED_EDGELS to determine the required array size for the M_CHAIN_... result types. This result type is not available if the target edges were not saved in the context (M_SAVE_TARGET_EDGES must be set to M_ENABLE). When retrieving a general result, this result type is not supported if the Model Finder context used was an M_GEOMETRIC_CONTROLLED type of context. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.															
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

When retrieving a general result or result(s) for a single occurrence, the ResultType parameter can be set

to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For an individual occurrence or a general result																		
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_CHAIN_ANGLE +	Retrieves the angle values of each edgel in all chains in the entire target or in the model occurrence. The angle is always returned in uncalibrated pixel units and is measured counter-clockwise in the image, from the positive X-axis towards the negative Y-axis of the image. Possible values range from 0 to 360 degrees and mapped in the range of 0 to 255 (that is, an 8-bit range). This result type is not available if the target edges were not saved in the context (M_SAVE_TARGET_EDGES must be set to M_ENABLE). When retrieving a general result, this result type is not supported if the Model Finder context used was an M_GEOMETRIC_CONTROLLED type of context. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.															
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_CHAIN_INDEX +	Retrieves the chain index of each edgel in all chains in the entire target, or in the model occurrence. The first chain in the target is identified as index 1, with increasing indices for subsequent chains. This result type is not available if the target edges were not saved in the context (M_SAVE_TARGET_EDGES must be set to M_ENABLE). When retrieving a general result, this result type is not supported if the Model Finder context used was an M_GEOMETRIC_CONTROLLED type of context. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_X +	Retrieves the X-coordinate of each edgel in all chains in the entire target or in the model occurrence. This result type is not available if the target edges were not saved in the context (M_SAVE_TARGET_EDGES must be set to M_ENABLE). When retrieving a general result, this result type is not supported if the Model Finder context used was an M_GEOMETRIC_CONTROLLED type of context. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_CHAIN_Y +	Retrieves the Y-coordinate of each edgel in all chains in the entire target or in the model occurrence. This result type is not available if the target edges were not saved in the context (M_SAVE_TARGET_EDGES must be set to M_ENABLE). When retrieving a general result, this result type is not supported if the Model Finder context used was an M_GEOMETRIC_CONTROLLED type of context. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_NUMBER_OF_CHAINED_EDGELS; M_CHAIN_INDEX; M_CHAIN_X; M_CHAIN_Y.

You can add one of the following values to the above-mentioned values to specify the edges for which to retrieve results.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which edges in the occurrence to consider																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_MODEL	Retrieves the edge information of the model transformed at the occurrence position.	c		e	f	g	a		i		j	k	l	m	n	b
M_TARGET	Retrieves the edge information of the target in the region of the occurrence. When using M_TARGET, you must not set the ResultIndex parameter to M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b

When retrieving a result for a single occurrence or for all occurrences, the ResultType parameter can be set to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a result of a single occurrence or all occurrences																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																	

M_ANGLE +	Retrieves the angle of the occurrence's reference axis (specified for the model using MmodControl() with M_REFERENCE_ANGLE), relative to the output coordinate system specified using MmodControl() with M_RESULT_OUTPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup. For an M_SHAPE_ELLIPSE type of Model Finder result buffer, the reference axis is the principal axis of the ellipse (its width), unless the reference axis is modified using MmodControl() with M_REFERENCE_ANGLE prior to calling MmodFind(). For an M_SHAPE_SEGMENT type of Model Finder result buffer, the orientation of the segment is dependent on its gradient, which is set using MmodControl() with M_SEGMENT_CONSISTENT_GRADIENT. The direction of the segment is orthogonal to its gradient.														
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_ASPECT_RATIO +	Retrieves the found aspect ratio of all occurrences. For an M_SHAPE_ELLIPSE or an M_SHAPE_RECTANGLE type of Model Finder result buffer, the returned aspect ratio is defined as being the ratio of the occurrence's aspect ratio divided by the model's. This result type is only available for an M_SHAPE_ELLIPSE, M_SHAPE_RECTANGLE, or M_DEFAULT type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM_LEFT_X +		Retrieves the X-coordinate of the bottom-left corner of the rectangle. This result type is only available for an M_SHAPE_RECTANGLE type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM_LEFT_Y +		Retrieves the Y-coordinate of the bottom-left corner of the rectangle. This result type is only available for an M_SHAPE_RECTANGLE type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM_RIGHT_X +		Retrieves the X-coordinate of the bottom-right corner of the rectangle. This result type is only available for an M_SHAPE_RECTANGLE type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM_RIGHT_Y +		Retrieves the Y-coordinate of the bottom-right corner of the rectangle. This result type is only available for an M_SHAPE_RECTANGLE type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_CENTER_X +	Retrieves the X-coordinate of the center of an occurrence of a circle, ellipse, rectangle, or segment model. This result type is only supported for an M_SHAPE_... type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Y +	Retrieves the Y-coordinate of the center of an occurrence of a circle, ellipse, rectangle, or segment model. This result type is only supported for an M_SHAPE_... type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_END_POS_X +	Retrieves the X-coordinate of the end point of an occurrence of a segment. This result type is only available for an M_SHAPE_SEGMENT type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_END_POS_Y +	Retrieves the Y-coordinate of the end point of an occurrence of a segment. This result type is only available for an M_SHAPE_SEGMENT type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_FIT_ERROR +		Retrieves the fit error as the average quadratic distance, in pixels or calibrated units, between the edges in the occurrence and the corresponding edges in the model. A perfect fit is represented as 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_INDEX +		Retrieves the index of the model that was found.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_COVERAGE +		Retrieves the model coverage of the occurrence. The model coverage is the percentage of the total length of the model's active edges found in the occurrence. 100% indicates that for every active edge in the model, a corresponding edge was found in the occurrence. Note that using a weighted mask with a model affects the calculation of the model coverage. To view the modified equation, refer to the Determining what is a match section of Chapter 8: Geometric Model Finder.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POLARITY +		Retrieves the polarity of the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the occurrences are a mixture of polarities. For more information, see MmodControl() with M_POLARITY and M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b

	M_REVERSE	Specifies that the polarity of occurrences is the reverse of that of the model. For more information, see MmodControl() with M_POLARITY and M_REVERSE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the polarity of occurrences is the same as that of the model. For more information, see MmodControl() with M_POLARITY and M_SAME.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X +		Retrieves the X-coordinate of the occurrence. This is the X-position of the model's reference axis transformed at the occurrence (MmodControl() with M_REFERENCE_X).	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y +		Retrieves the Y-coordinate of the occurrence. This is the Y-position of the model's reference axis transformed at the occurrence (MmodControl() with M_REFERENCE_Y).	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SCALE +		Retrieves the scale of the occurrence. For an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder result buffer, the returned scale is defined as the occurrence's width divided by the model's width. For an M_SHAPE_SEGMENT type of Model Finder result buffer, the returned scale is defined as the occurrence's length divided by the model's length.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_SCORE +	Retrieves the score of the occurrence (as a percentage). The score is calculated as follows: Score = Model coverage x (1 - Fit error weighting factor * Normalized fit error) Note: the normalized fit error is the fit error (M_FIT_ERROR) converted to a number between 0.0-1.0 for an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, while for an M_SHAPE_CIRCLE or M_SHAPE_CIRCLE type of Model Finder context, the normalized fit error is influenced by the value of M_SAGITTA_TOLERANCE and is not limited to being between 0.0 and 1.0. For an M_SHAPE_RECTANGLE or M_SHAPE_SEGMENT type of Model Finder context, the normalized fit error is influenced by the value of M_DEVIATION_TOLERANCE and is not limited to being between 0.0 and 1.0. For all Model Finder context types, the terms within the square brackets result in a number between 0.0 and 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SCORE_FIT +	Retrieves the fit score of the occurrence (as a percentage). The score is calculated as follows: Fit Score = 1 - Normalized fit error This result type is only supported for a model defined in an M_SHAPE_... type of Model Finder context. For other types of models see the target score (M_SCORE_TARGET).	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_SCORE_TARGET +	Retrieves the target score of the occurrence. The target score is calculated as follows: Target Score = Target coverage x (1 - Fit error weighting factor * Normalized fit error) Note that the normalized fit error (M_FIT_ERROR) is the fit error converted to a number between 0.0 and 1.0. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SEGMENT_CONSISTENT_GRADIENT +	Retrieves whether the gradients of the segment model occurrences are consistent.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the occurrences do not have consistent gradients.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the occurrences have consistent gradients.	c		e	f	g	a		i		j	k	l	m	n	b
M_START_POS_X +	Retrieves the X-coordinate of the start point of an occurrence of a segment. This result type is only available for an M_SHAPE_SEGMENT type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_START_POS_Y +	Retrieves the Y-coordinate of the start point of an occurrence of a segment. This result type is only available for an M_SHAPE_SEGMENT type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TARGET_COVERAGE +	Retrieves the target coverage of the model. The target coverage is a percentage of the total length of edges present within the occurrence's bounding box, corresponding to the model's active edges. Lower scores indicate that features or edges found in the target (result occurrence) are not present in the model. Note that using a weighted mask with a model affects the calculation of the target coverage. To view the modified equation, refer to the Determining what is a match section of Chapter 8: Geometric Model Finder. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP_LEFT_X +	Retrieves the X-coordinate of the top-left corner of the rectangle. This result type is only available for an M_SHAPE_RECTANGLE type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP_LEFT_Y +	Retrieves the Y-coordinate of the top-left corner of the rectangle. This result type is only available for an M_SHAPE_RECTANGLE type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP_RIGHT_X +		Retrieves the X-coordinate of the top-right corner of the rectangle. This result type is only available for an M_SHAPE_RECTANGLE type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP_RIGHT_Y +		Retrieves the Y-coordinate of the top-right corner of the rectangle. This result type is only available for an M_SHAPE_RECTANGLE type of Model Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_LABEL +		Retrieves the user label of the model that was found.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_LABEL	Specifies that no user label is associated with the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the user label of the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b

When retrieving a result for a single occurrence or for all occurrences, the ResultType parameter can be set to one of the following values. The following values are not available for all model types. For one of these values to be available for a result occurrence, the result occurrence must be of the appropriate type. Use M_AVAILABLE to ensure that the value can be used with the occurrence. If the ResultIndex parameter is set to M_ALL, all the occurrences must support the result type, otherwise an error is generated.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a result of a single occurrence or all occurrences (with constraints)

ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_CORNER_RADIUS +	Retrieves the radius used to round the corners of the models. This control type is only valid for models of type M_CROSS, M_DIAMOND, M_RECTANGLE, M_SQUARE, and M_TRIANGLE. This result type is not available for an M_SHAPE_... type of Model Finder result buffer.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_HEIGHT +	Retrieves the height of the shape. This result type is only available for M_CROSS, M_DIAMOND, M_ELLIPSE, M_RECTANGLE and M_TRIANGLE.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_HORIZONTAL_THICKNESS +	Retrieves the horizontal thickness of the cross. This result type is only available for M_CROSS.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_INNER_RADIUS +	Retrieves the inner radius of the ring. This result type is only available for M_RING.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH +	Retrieves the length of the segment. This result type is only available for M_SEGMENT.	c		e	f	g	a		i		j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_OUTER_RADIUS +	Retrieves the outer radius of the ring. This result type is only available for M_RING.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS +	Retrieves the radius of the circle. This result type is only available for M_CIRCLE.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_VERTICAL_THICKNESS +	Retrieves the vertical thickness of the cross. This result type is only available for M_CROSS.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_WIDTH +	Retrieves the width of the shape. This result type is only available for M_CROSS, M_DIAMOND, M_ELLIPSE, M_RECTANGLE, M_SQUARE and M_TRIANGLE.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve a transformation coefficient for a single occurrence or for all occurrences, the ResultType parameter can be set to one of the following values. These coefficients allow you to convert coordinates in the model coordinate system to the corresponding coordinates in the target coordinate system for that occurrence (or vice versa). These coefficients handle variations in scale, translation, and angle. If the model is calibrated, these coefficients are given for the real world.

Use the following equations:

$$x_d = A x_s + B y_s + C$$

$$y_d = E x_s + F y_s + D$$

For models that do not have an M_ASPECT_RATIO result type, note the following:

$E = -B$

$F = A$

where A, B, C, D, E, and F are the transformation coefficients (forward or reverse); x_s and y_s specify the source coordinates (with respect to the origin of the model coordinate system for a forward transformation or target coordinate system for a reverse transformation); and, x_d and y_d are the destination coordinates (with respect to the origin of the target coordinate system for a forward transformation or model coordinate system for a reverse transformation).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving a transformation coefficient																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																	
M_A_FORWARD +	Retrieves the forward transformation coefficient A for the occurrence.																
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_A_REVERSE +	Retrieves the reverse transformation coefficient A for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_B_FORWARD +	Retrieves the forward transformation coefficient B for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_B_REVERSE +	Retrieves the reverse transformation coefficient B for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_C_FORWARD +	Retrieves the forward transformation coefficient C for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_C_REVERSE +	Retrieves the reverse transformation coefficient C for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_D_FORWARD +	Retrieves the forward transformation coefficient D for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_D_REVERSE +	Retrieves the reverse transformation coefficient D for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_E_FORWARD +	Retrieves the forward transformation coefficient E for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_E_REVERSE +	Retrieves the reverse transformation coefficient E for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_F_FORWARD +	Retrieves the forward transformation coefficient F for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_F_REVERSE +	Retrieves the reverse transformation coefficient F for the occurrence.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for
Tables:

- For a general result
- For an individual occurrence, all occurrences, or a general result
- For an individual occurrence or a general result
- For a result of a single occurrence or all occurrences
- For a result of a single occurrence or all occurrences (with constraints)
- For retrieving a transformation coefficient

the values listed in all the parameter association tables.

You can add the following value to the above-mentioned values to specify whether a result is available.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining whether results are available																	
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ResultArrayPtr - Possible values returned																
M_AVAILABLE		Retrieves whether the requested result type is available for retrieval.															
	ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* **|** *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the requested result type is not available.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the requested result type is available.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for
Tables:

- For an individual occurrence, all occurrences, or a general result
- For an individual occurrence or a general result
- For a result of a single occurrence or all occurrences
- For a result of a single occurrence or all occurrences (with constraints)
- For retrieving a transformation coefficient

the values listed in all the parameter association tables except **For a general result**.

You can add the following value to the above-mentioned values to determine the required array size (number of elements) to store the returned values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining the required array size (number of elements) to store the returned values																	
ResultType combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_NB_ELEMENTS	Retrieves the required array size (number of elements) to store the returned values.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For a general result
- For an individual occurrence, all occurrences, or a general result
- For an individual occurrence or a general result
- For a result of a single occurrence or all occurrences
- For a result of a single occurrence or all occurrences (with constraints)
- For retrieving a transformation coefficient

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested results to the required data type.

> Note: In C++, when using a standard vector (std::vector) instead of an array, the values listed in this combination table must not be used. If you need to use a specific data type, instantiate the vector with the necessary data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested results to an *AIL_DOUBLE*.															
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* **|** *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_ID	Casts the requested results to an *AIL_ID*.	c		e	f	g	a		i		j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]* **|** *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested results to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]* **|** *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested results to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]* **|** *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested results to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]* **|** *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodInquire

Synopsis

Inquire information about a specified Model Finder context, a specified model, or a specified result buffer.

Syntax

```
AIL_INT MmodInquire(
    AIL_ID ContextOrResultId, //in
    AIL_INT Index,            //in
    AIL_INT64 InquireType,    //in
    void *UserVarPtr          //out
)
```

Description

This function inquires information about a specified Model Finder context, the models contained therein, or a specified Model Finder result buffer. Note that Model Finder result buffer values can be obtained with MmodGetResult().

In general, returned values (for example, position and width) are in the following units:

- For a non-synthetic model, returned values are in the pixel coordinate system, even if the target is calibrated. The only exception is when inquiring about angles (for example, M_REFERENCE_ANGLE); in these cases, the angles values are in the relative coordinate system if the target is calibrated.
- For a synthetic model in an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED context, returned values are in the user-defined units for the model; their relationship to pixels is defined using MmodControl() with M_PIXEL_SCALE. If the target is not calibrated, M_PIXEL_SCALE is used for the match. If the target is calibrated, model units are used; note that these should be the same as the calibrated units. In a few cases, instead of returning values in model units, values are returned in pixel units (for example, M_POSITION_...); these cases are clearly indicated.
- For a model in an M_SHAPE_... context, returned values are in model units, which in this case are the same as the target units. If the target is calibrated, the model's settings will be interpreted in world units. If the target is not calibrated, the model's settings will be interpreted in pixel units.

If an inquire type is set to M_DEFAULT, **MmodInquire**() will return M_DEFAULT. To inquire its default value, add M_DEFAULT to the inquire type.

Parameters

ContextOrResultId

Specifies the Model Finder context or Model Finder result buffer about which to inquire information. The Model Finder context or result buffer must have been previously allocated on the required system using MmodAlloc() or MmodAllocResult(), respectively.

Index

Specifies that information will be inquired about the Model Finder context, an individual model, or a Model Finder result buffer. This parameter should be set to one of the following values:

For specifying a context, model, or result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default. If a Model Finder context is specified, this parameter inquires information about model index 0. If a result buffer is specified, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Inquires information about the Model Finder context, which has been set using the ContextOrResultId parameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Inquires general information about the Model Finder result buffer, which has been set using the ContextOrResultId parameter.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *IndexValue* < M_NUMBER_MODELS	Specifies the index of the individual model to inquire, if ContextOrResultId is set to a Model Finder context. To retrieve the index of a model from its user label, set this parameter to the user label and set the inquire type to M_INDEX_FROM_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b
LabelValue	Specifies the user label of the individual model to inquire, if ContextOrResultId is set to a Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b

InquireType

Specifies the type of setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]
- array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]
- array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]

- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **MmodInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For inquiring about the number of occurrences to find for the entire context or an individual model specified using its index
- For inquiring about the context or result buffer
- For inquiring about the context
- For inquiring about a model using its index
- For inquiring about a model using its user label
- For a result buffer

For M_CONTEXT or an individual model specified using its index, the InquireType parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the number of occurrences to find for the entire context or an individual model specified using its index																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_NUMBER +		Inquires the number of models for which to search. For M_CONTEXT, this setting returns the maximum number of occurrences of all models to find in the target. For a model, this setting returns the maximum number of occurrences of the specified model to find in the target. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies to find all occurrences. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0	Specifies the number of occurrences. INFO	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about the system on which the Model Finder context or result buffer has been allocated, set the InquireType parameter to the value below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the context or result buffer																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_OWNER_SYSTEM +		Inquires the identifier of the system on which the Model Finder context or result buffer was allocated. SET SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
	System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

For M_CONTEXT, the InquireType parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the context																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_ACCURACY +		Inquires the search accuracy. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies high accuracy.	c		e	f	g	a		i		j	k	l	m	n	b

	M_LOW	Specifies low accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEDIUM	Specifies medium accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE_EDGELS +		Inquires the degree to which Aurora Imaging Library processes active edgels in the target. This inquire type is only supported for a geometric Model Finder context (MmodAlloc() with M_GEOMETRIC). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 100.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the degree to which Aurora Imaging Library processes target edgels, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ASPECT_RATIO +		Inquires the aspect ratio applied to the target before starting the search. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.5 <= *Value* <= 2.0	Specifies the pixel width/pixel height of the target.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT_TYPE +		Inquires the type of search context allocated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_GEOMETRIC	Specifies that the Model Finder context uses a general geometric search algorithm. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GEOMETRIC_CONTROLLED	Specifies that the Model Finder context uses a controlled geometric search algorithm. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHAPE_CIRCLE	Specifies that the Model Finder context uses a circular model search algorithm. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHAPE_ELLIPSE	Specifies that the Model Finder context uses an elliptical model search algorithm. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHAPE_RECTANGLE	Specifies that the Model Finder context uses a rectangular model search algorithm. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHAPE_SEGMENT	Specifies that the Model Finder context uses a segment model search algorithm. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DETAIL_LEVEL +		Inquires the level of details to extract from the model source and target image. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies a high level of detail.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEDIUM	Specifies a medium level of detail.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies a very high level of detail.	c		e	f	g	a		i		j	k	l	m	n	b
M_EDGE_ACCURACY +		Inquires the edge accuracy used when calculating the edges. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies a high level of edge accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies a very high level of edge accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_LEVEL +		Inquires the resolution level for the initial stage of the search. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies that the first resolution is determined automatically.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <= 7	Specifies the resolution level. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LAST_LEVEL +		Inquires the resolution level for the final stage of the search. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO	Specifies that the last resolution is determined automatically.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <= 7	Specifies the resolution level. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_MODIFICATION_COUNT +	Inquires the current value of the modification counter. The modification counter is increased by one each time settings for the context are modified. Although you cannot identify the modification counter's contents, you can compare them throughout your application to know if the context has been altered. If the modification counter has changed you can, for example, prompt the user to save before closing the application.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the current value of the modification counter.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_MODELS +	Inquires the number of models in the context.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the number of models.	c		e	f	g	a		i		j	k	l	m	n	b
M_PREPROCESSED +	Inquires whether the Model Finder context is preprocessed. The context must be preprocessed (using MmodPreprocess()) before calling MmodFind(). After certain settings of the Model Finder context are changed with MmodControl(), or after a model is added or removed with MmodDefine(), this inquire type will indicate that the context is no longer in its preprocessed state.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the context is not preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRUE	Specifies that the context is preprocessed.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESOLUTION_COARSENESS_LEVEL +		Inquires the resolution of the edge approximation. This inquire type is only supported for a model defined in an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the resolution. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_TARGET_EDGES +		Inquires whether the target edges in the result will be saved. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that target edges cannot be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that target edges can be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_ANGLE_RANGE +		Inquires whether to perform calculations specific to angular-range search strategies. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that calculations specific to angular-range search strategies are disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that calculations specific to angular-range search strategies are enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_POSITION_RANGE +		Inquires whether to perform calculations specific to position-range search strategies. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that calculations specific to position-range search strategies are disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that calculations specific to position-range search strategies are enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_SCALE_RANGE +		Inquires whether to perform calculations specific to scale-range search strategies. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that calculations specific to scale-range search strategies are disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that calculations specific to scale-range search strategies are enabled. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_SHARED_EDGES +		Inquires whether multiple occurrences can share edges. This inquire type is not supported for a model defined in an M_SHAPE_CIRCLE, an M_SHAPE_ELLIPSE, or an M_SHAPE_SEGMENT type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the sharing of edges. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to enable the sharing of edges.	c		e	f	g	a		i		j	k	l	m	n	b
M_SMOOTHNESS +		Inquires the degree of smoothness (denoising) applied to the model images and the target images during edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the smoothness value applied to the images. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SPEED +		Inquires the search speed. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies a high speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LOW	Specifies a low speed.	c		e	f	g	a		i		j	k	l	m	n	b

M_MEDIUM	Specifies a medium speed.	c		e	f	g	a		i		j	k	l	m	n	b
M_VERY_HIGH	Specifies a very high speed.	c		e	f	g	a		i		j	k	l	m	n	b
M_TARGET_CACHING +	Inquires whether target caching is enabled. This inquire type is not supported for an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies to disable target caching. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to enable target caching in the result.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT +	Inquires the maximum search time for MmodFind(), in msec. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 2000.0 msec.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies an infinite amount of search time.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the maximum search time, in msecs.	c		e	f	g	a		i		j	k	l	m	n	b

For inquiring about an individual model specified using its index, the InquireType parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a model using its index																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

	UserVarPtr - Possible values returned															
M_ACCEPTANCE +		Inquires the acceptance level for the score. SET														
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 60.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies an acceptable score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCEPTANCE_TARGET +		Inquires the acceptance level for the target score. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies an acceptable target score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ALLOC_OFFSET_X +		Inquires the X-offset of the model in the model source image. This is only supported for image-type or Edge Finder-type models. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-offset of the model, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALLOC_OFFSET_Y +		Inquires the Y-offset of the model in the model source image. This is only supported for image-type or Edge Finder-type models. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-offset of the model, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALLOC_SIGN +		Inquires whether the data range of the model image is signed or unsigned.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_SIGNED	Specifies that the data is signed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_UNSIGNED	Specifies that the data is unsigned. Synthetic, Model Finder-type and merge-type models will always return M_UNSIGNED.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALLOC_SIZE_BAND +		Inquires the number of bands of the model image. When drawing the model image, use M_ALLOC_SIZE_BAND to establish the required number of bands for the destination buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	1	Specifies a 1-band buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	3	Specifies a 3-band buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALLOC_SIZE_BIT +		Inquires the depth per band, in bits, of the model image.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	1	Specifies that the data depth is 1 bit per band.	c		e	f	g	a		i		j	k	l	m	n	b
	8	Specifies that the data depth is 8 bits per band.	c		e	f	g	a		i		j	k	l	m	n	b
	16	Specifies that the data depth is 16 bits per band.	c		e	f	g	a		i		j	k	l	m	n	b

	32	Specifies that the data depth is 32 bits per band.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALLOC_SIZE_X +		Inquires the width of the smallest buffer in which to draw the model box (the model and its margins). For a model of an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, this value also determines the mask size. For a synthetic model of an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, this value is affected by M_PIXEL_SCALE and model calibration. For a synthetic model of an M_SHAPE_... type of Model Finder context, this value is determined by the smallest buffer that can contain the width of the model, and the margin that adds 10% of the width of the bounding box's active edges to the left and right edges of the model box. Note that the width of the model will be interpreted as if it had been specified in pixel units (even if they were not). If they were specified assuming a calibrated target, the returned size might not be very meaningful. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 16	Specifies the width of the model (in X), in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

M_ALLOC_SIZE_Y +	Inquires the height of the smallest buffer in which to draw the model box (the model and its margins). For a model of an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, this value also determines the mask size. For a synthetic model of an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context, this value is affected by M_PIXEL_SCALE and model calibration. For a synthetic model of an M_SHAPE_... type of Model Finder context, this value is determined by the smallest buffer that can contain the height of the model, and the margin that adds 10% of the height of the bounding box's active edges to the top and bottom edges of the model box. Note that the height of the model will be interpreted as if it had been specified in pixel units (even if they were not). If they were specified assuming a calibrated target, the returned size might not be very meaningful. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 16	Specifies the height of the model (in Y), in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALLOC_TYPE +	Inquires the data type and depth of the model image.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_FLOAT + 32	Specifies 32-bit floating-point data.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIGNED + 8	Specifies 8-bit signed data.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIGNED + 16	Specifies 16-bit signed data.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIGNED + 32	Specifies 32-bit signed data.	c		e	f	g	a		i		j	k	l	m	n	b

	M_UNSIGNED + 1	Specifies 1-bit unsigned data.	c		e	f	g	a		i		j	k	l	m	n	b
	M_UNSIGNED + 8	Specifies 8-bit unsigned data.	c		e	f	g	a		i		j	k	l	m	n	b
	M_UNSIGNED + 16	Specifies 16-bit unsigned data.	c		e	f	g	a		i		j	k	l	m	n	b
	M_UNSIGNED + 32	Specifies 32-bit unsigned data.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE +		Inquires the nominal search angle; this is the angle at which you expect to find the model's reference axis (specified using M_REFERENCE_ANGLE) in the target. For an ellipse or rectangle model defined in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context, the model's reference axis is, by default, aligned with its principle axis (its width). This inquire type is not supported for a model defined in an M_SHAPE_CIRCLE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0 °.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the nominal search angle, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_DELTA_NEG +		Inquires the lower limit of the angular range, relative to the nominal search angle. This inquire type is not supported for a model defined in an M_SHAPE_CIRCLE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 180.0 °.	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 <= *Value* <= 180.0	Specifies the lower limit of the angular range, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_DELTA_POS +		Inquires the upper limit of the angular range, relative to the nominal search angle. This inquire type is not supported for a model defined in an M_SHAPE_CIRCLE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 180.0 °.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 180.0	Specifies the upper limit of the angular range, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_MULTIPLE_RANGE +		Inquires whether to search for models in multiple angular ranges. This inquire type is only supported for a model in an M_SHAPE_RECTANGLE or M_SHAPE_SEGMENT type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_STEP_180.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that searching in multiple angular ranges is disabled. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_STEP_90	Specifies to search for models at angular ranges at 90 degree steps from the nominal search angle. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_STEP_180	Specifies to search for models at a 180 degree step. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_ANGLE_REGION +		Sets the angle direction of the search region defined by the M_POSITION_... controls. This control type is only valid for M_SHAPE_... models. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0°.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the search angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_ASSOCIATED_CALIBRATION +		Inquires the camera calibration context associated with the specified model. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL	Specifies that there is no camera calibration context associated with the model.	c		e	f	g	a		i		j	k	l	m	n	b
	Calibration context identifier	Specifies the camera calibration context that is associated with the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_MARGIN_BOTTOM +		Inquires the margin at the bottom of the bounding box of the model's active edges, in model units. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these contexts, the bottom margin is always 10.0% of the height of the bounding box of the model's active edges. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 10.0% of the height of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_MARGIN_LEFT +		Inquires the margin at the left of the bounding box of the model's active edges, in model units. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these contexts, the left margin is always 10.0% of the width of the bounding box of the model's active edges. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0% of the width of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_MARGIN_RIGHT +		Inquires the margin at the right of the bounding box of the model's active edges, in model units. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these contexts, the right margin is always 10.0% of the width of the bounding box of the model's active edges. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0% of the width of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

M_BOX_MARGIN_TOP +		Inquires the margin at the top of the bounding box of the model's active edges, in model units. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. For these contexts, the top margin is always 10.0% of the height of the bounding box of the model's active edges. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0% of the height of the bounding box of the model's active edges.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the margin, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_OFFSET_X +		Inquires the X-offset of the top-left corner of the model box from the model origin. For predefined shape models, the origin of the model is the center of the shape. For CAD-file models, the origin is the same as the one defined in the CAD file. The model box is defined by the bounding box of the model's active edges plus the specified margins. You can inquire the margins using M_BOX_MARGIN_.... M_BOX_OFFSET_X is updated when the margins that define the model box are changed. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-offset, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

M_BOX_OFFSET_Y +		Inquires the Y-offset of the top-left corner of the model box from the model origin. For predefined shape models, the origin of the model is the center of the shape. For CAD-file models, the origin is the same as the one defined in the CAD file. The model box is defined by the bounding box of the model's active edges plus the specified margins. You can inquire the margins using M_BOX_MARGIN_.... M_BOX_OFFSET_Y is updated when the margins that define the model box are changed. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-offset, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_SIZE_X +		Inquires the size along the X-axis of the model box. The model box is defined by the bounding box of the model's active edges plus the specified margins. You can inquire the margins using M_BOX_MARGIN_.... You cannot set the size of the model box; however, you can set the size of the model's bounding box when defining the model using MmodDefine(). You can also set the size of the margins using MmodControl() with M_BOX_MARGIN_.... This inquire type is only valid for synthetic models. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value	Specifies the X-size, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_SIZE_Y +		Inquires the size along the Y-axis of the model box. The model box is defined by the bounding box of the model's active edges plus the specified margins. You can inquire the margins using M_BOX_MARGIN_.... You cannot set the size of the model box; however, you can set the size of the model's bounding box when defining the model using MmodDefine(). You can also set the size of the margins using MmodControl() with M_BOX_MARGIN_.... This inquire type is only valid for synthetic models. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-size, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_CAD_Y_AXIS +		Inquires the direction of the Y-axis for a CAD-type model. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FLIP	Specifies to flip the Y-axis for the model so that the Y-axis is positive going down.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_FLIP	Specifies not to flip the Y-axis for the model.	c		e	f	g	a		i		j	k	l	m	n	b

M_CERTAINTY +		Inquires the certainty level for the score. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 90.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the certainty level for the score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CERTAINTY_TARGET +		Inquires the certainty level for the target score. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the certainty level for the target score, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_ANGLE +		Inquires the angle value of each edgel in the chains composing the model's active edges. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the angle value of the edgel.	c		e	f	g	a		i		j	k	l	m	n	b

M_CHAIN_INDEX +	Inquires the indices which differentiate each active edge's chain of edgels, for each edgel within the model. The first chain is identified as index 1, with each subsequent chain's index incremented by 1. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the index of the chain.	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_X +	Inquires the X-coordinates, in pixels (with subpixel accuracy), of each edgel in the chains composing the model's active edges. Coordinates are returned for all the active edge chains contained within the model and are relative to the model origin. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate of the edgel.	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_Y +	Inquires the Y-coordinates, in pixels (with subpixel accuracy), of each edgel in the chains composing the model's active edges. Coordinates are returned for all the active edge chains contained within the model and are relative to the model origin. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-coordinate of the edgel.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_RADIUS +		Inquires the radius used to round corners of the model. This is only valid for models of type M_RECTANGLE, M_SQUARE, M_DIAMOND, M_TRIANGLE, and M_CROSS. Attempting to call this value for any other model type will generate an error. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the radius, in model units. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COVERAGE_MAX +		Inquires the maximum expected model coverage. This inquire type is only supported for a model defined in an M_SHAPE_CIRCLE or M_SHAPE_ELLIPSE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 100.0	Specifies the maximum expected model coverage.	c		e	f	g	a		i		j	k	l	m	n	b

M_COVERAGE_MIN +		Inquires the minimum expected model coverage. This inquire type is only supported for a model defined in an M_SHAPE_SEGMENT type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 40.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 100.0	Specifies the minimum expected model coverage.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEVIATION_TOLERANCE +		Inquires the tolerance for finding deformed rectangles or segments, for a model defined in an M_SHAPE_RECTANGLE or M_SHAPE_SEGMENT type of Model Finder context, respectively, given the other specified Model Finder constraints. This inquire type is only supported for M_RECTANGLE or M_SEGMENT types of models in an M_SHAPE_RECTANGLE or M_SHAPE_SEGMENT type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 25.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the tolerance as a percentage of the allowable deformation of a rectangle or segment, given the other Model Finder constraints. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_ELLIPSE_TOLERANCE_TO_CIRCLE +		Inquires the tolerance for approximating the aspect ratio of an ellipse occurrence to 1 (that is, reporting that the ellipse model matched a circle). M_ELLIPSE_TOLERANCE_TO_CIRCLE is only supported for a model defined in an M_SHAPE_ELLIPSE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.001.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 0.5	Specifies the tolerance applied to the ellipse occurrence's aspect ratio. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_ERROR_WEIGHTING_FACTOR +		Inquires the contribution of the fit error in the score and target score calculation. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 25.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the fit error weighting factor, as a percentage. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_SCORE_MIN +		Inquires the minimum expected occurrence fit score. This inquire type is only supported for a model defined in an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the minimum expected occurrence fit score.	c		e	f	g	a		i		j	k	l	m	n	b

M_FOREGROUND_VALUE +	Inquires the foreground value of the model. The foreground is the interior of the shape. This is only valid for predefined shape models. CAD-type models do not have a polarity. For these model types, M_ANY will be returned. SET SET SET SET SET SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ANY	Specifies that the model has no specific polarity. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FOREGROUND_BLACK	Specifies that black is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_FOREGROUND_WHITE	Specifies that white is the foreground color of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_HAS_DONT_CARE_MASK +	Inquires whether the model has an M_DONT_CARE mask applied to it.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that an M_DONT_CARE mask has not been applied to the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that an M_DONT_CARE mask has been applied to the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_HAS_FLAT_REGIONS_MASK +	Inquires whether the model has an M_FLAT_REGIONS mask applied to it.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that an M_FLAT_REGIONS mask has not been applied to the model.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRUE	Specifies that an M_FLAT_REGIONS mask has been applied to the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_HAS_WEIGHT_REGIONS_MASK +		Inquires whether the model has an M_WEIGHT_REGIONS mask applied to it.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that an M_WEIGHT_REGIONS mask has not been applied to the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that an M_WEIGHT_REGIONS mask has been applied to the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_HEIGHT +		Inquires the height of the shape in the model. This is only valid for a model of type M_CROSS, M_DIAMOND, M_ELLIPSE, M_RECTANGLE, and M_TRIANGLE. SET SET SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the height of the shape, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_HORIZONTAL_THICKNESS +		Inquires the horizontal thickness of the cross in the model. This is only valid for a model of type M_CROSS. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the horizontal thickness of the cross, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_INNER_RADIUS +		Inquires the inner radius of the ring in the model. This is only valid for a model of type M_RING. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the inner radius of the ring, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH +		Inquires the length of the segment in the model. This is only valid for a model of type M_SEGMENT. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the length of the segment, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_ANGLE +		Inquires the minimum angular separation required for two occurrences to be considered distinct matches. This value is an absolute angle value. This inquire type is not supported for a model defined in an M_SHAPE_CIRCLE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Disables the minimum angle separation criteria. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 180.0	Specifies the minimum angular separation required, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_ASPECT_RATIO +		Inquires the minimum separation required in aspect ratios, for two occurrences to be considered distinct matches. This value is specified as an aspect ratio factor. This inquire type is only supported for a model in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the minimum aspect ratio separation criteria. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 < *Value* <= 4.0	Specifies the criteria for minimum separation of aspect ratios.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_SCALE +		Inquires the minimum separation required in scale for two occurrences to be considered distinct matches. This value is a scale factor. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the minimum scale separation criteria. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 < *Value* <= 4.0	Specifies the criteria for minimum separation in scale.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_X +		Inquires the minimum separation required along the X-axis for two occurrences to be considered distinct matches. This value is a percentage of the model's width at M_SCALE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the minimum separation in X criteria. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	Value	Specifies the minimum separation as a percentage of the model's width at M_SCALE. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SEPARATION_Y +		Inquires the minimum separation required along the Y-axis for two occurrences to be considered distinct matches. This value is a percentage of the model's height at M_SCALE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the minimum separation in Y criteria. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the minimum separation as a percentage of the model's height at M_SCALE. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_SIDE_COVERAGE +		Inquires the minimum coverage on each of the sides of a rectangular model. Note that at least a portion of each side must be visible. This inquire type is only supported for a model in an M_SHAPE_RECTANGLE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 40.0.	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 <= *Value* <= 100.0	Specifies the minimum side coverage.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODEL_ASPECT_RATIO +		Inquires the nominal aspect ratio factor for the model. This inquire type is only supported for a model in an M_SHAPE_ELLIPSE or an M_SHAPE_RECTANGLE type of Model Finder context. For other types of Model Finder contexts, see M_ASPECT_RATIO. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_ASPECT_RATIO	Specifies the minimum possible value for this control type.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the value of the nominal aspect ratio for the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_ASPECT_RATIO_MAX_FACTOR +		Inquires the factor used to determine the upper limit (maximum permitted aspect ratio) of the model's aspect ratio. This inquire type is only supported for a model in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.2.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies that the maximum factor for the aspect ratio of a model is infinite.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1.0	Specifies the factor that determines the maximum aspect ratio for a model.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODEL_ASPECT_RATIO_MIN_FACTOR +		Inquires he factor used to determine the lower limit (minimum permitted aspect ratio) of the model's aspect ratio. This inquire type is only supported for a model in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.8.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_ASPECT_RATIO	Specifies the minimum factor for the aspect ratio of a model.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 1.0	Specifies the factor that determines the minimum aspect ratio for a model.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_IMAGE_ORIGIN_X +		Inquires the X-coordinate of the model origin, relative to the top-left corner of the smallest buffer that can contain the model box (the model and its margins). The size of this buffer can be inquired using M_ALLOC_SIZE_X and M_ALLOC_SIZE_Y. This inquire type can be useful if you want to draw a shape at the center of the model origin (the center of the predefined-shape model). To inquire the X-coordinate of the reference axis origin relative to the model origin, use M_REFERENCE_X. This inquire type is only supported for a model defined in an M_SHAPE_... type of Model Finder context. Note that the model origin for predefined-shape models is the center of the shape.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0	Specifies the X-coordinate of the model origin, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_IMAGE_ORIGIN_Y +		Inquires the Y-coordinate of the model origin, relative to the top-left corner of the smallest buffer that can contain the model box (the model and its margins). The size of this buffer can be inquired using M_ALLOC_SIZE_X and M_ALLOC_SIZE_Y. This inquire type can be useful if you want to draw a shape at the center of the model origin (the center of the predefined-shape model). To inquire the Y-coordinate of the reference axis origin relative to the model origin, use M_REFERENCE_X. This inquire type is only supported for a model defined in an M_SHAPE_... type of Model Finder context. Note that the model origin for predefined-shape models is the center of the shape.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the Y-coordinate of the model origin, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_TYPE +		Inquires the type of model. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AUTO_DEFINE	Specifies a unique model whose features are extracted from an automatically selected region of the model source image. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE	Specifies a predefined circle as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CROSS	Specifies a predefined cross as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_DIAMOND	Specifies a predefined diamond as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DXF_FILE	Specifies a model defined from the entities in the specified CAD DXF file. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGE_RESULT	Specifies a model defined from the results of an edge extraction performed using Edge Finder. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE	Specifies a predefined ellipse as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_IMAGE	Specifies a model defined from the specified model source image. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_MERGE_MODEL	Specifies a model defined from two other models. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_MOD_RESULT	Specifies a model defined from a Model Finder result buffer. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RECTANGLE	Specifies a predefined rectangle as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RING	Specifies a predefined ring as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SEGMENT	Specifies a predefined segment as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SQUARE	Specifies a predefined square as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRIANGLE	Specifies a predefined triangle as the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_NUMBER_OF_CHAINED_EDGELS +		Inquires the total number of chained edgels in the active edges of the model. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the number of chained edgels.	c		e	f	g	a		i		j	k	l	m	n	b
M_ORIGINAL_X +		Inquires the X-coordinate of the model's reference axis origin, relative to the top-left corner of the model source image. This is only valid for image-type and Edge Finder-type models. To inquire the X-coordinate of the reference axis origin relative to the model origin, use M_REFERENCE_X. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. Note that the model origin for image-type and Edge Finder-type models is the center of the upper left pixel of the model image.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-coordinate of the model's reference axis origin.	c		e	f	g	a		i		j	k	l	m	n	b

M_ORIGINAL_Y +		Inquires the Y-coordinate of the model's reference axis origin, relative to the top-left corner of the model source image This is only valid for image-type and Edge Finder-type models. To inquire the Y-coordinate of the reference axis origin relative to the model origin, use M_REFERENCE_Y. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context. Note that the model origin for image-type and Edge Finder-type models is the center of the upper left pixel of the model image.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-coordinate of the model's reference axis origin.	c		e	f	g	a		i		j	k	l	m	n	b
M_OUTER_RADIUS +		Inquires the outer radius of the ring in the model. This is only valid for a model of type M_RING. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the outer radius of the ring, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

M_PIXEL_SCALE +		Inquires the pixel scale of the model, if it is a synthetic model. M_PIXEL_SCALE is used for the match if the target is not calibrated; if the target is calibrated, this value is not used for the match nor for the returned results. Regardless if calibrated, this value will affect M_ALLOC_SIZE_X and M_ALLOC_SIZE_Y and it will also be used for draw operations. This is only valid for synthetic models. This inquire type is not supported for models defined in an M_SHAPE_... type of Model Finder context. For these types of contexts, model units are interpreted to be in the units of the target. If the target is calibrated, the units are interpreted to be in calibrated units. If the target is not, the units are interpreted to be in pixel units. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value of the pixel scale. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the pixel scale. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_POLARITY +		Inquires the expected polarity of occurrences relative to the model. This inquire type is not supported for a model defined in an M_SHAPE_SEGMENT type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the occurrences can be a mixture of polarities.	c		e	f	g	a		i		j	k	l	m	n	b

	M_REVERSE	Specifies that the polarity of occurrences is the reverse of that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the polarity of occurrences is the same as that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME_OR_REVERSE	Specifies that the polarity of occurrences can be either the same or the reverse of that of the model.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_DELTA_NEG_X +		Inquires the position range, in the negative X-direction relative to the nominal search position. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies the position range as the entire image plane in the negative X-direction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the position range's negative X-offset, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the offset in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_DELTA_NEG_Y +		Inquires the position range, in the negative Y-direction relative to the nominal search position. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies the position range as the entire image plane in the negative Y-direction. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	Value >= 0.0	Specifies the position range's negative Y-offset, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the offset in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_DELTA_POS_X +		Inquires the position range, in the positive X-direction relative to the nominal search position. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies the position range as the entire image plane in the positive X-direction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the position range's positive X-offset, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the offset in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_DELTA_POS_Y +		Inquires the position range, in the positive Y-direction relative to the nominal search position. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies the position range as the entire image plane in the positive Y-direction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the position range's positive Y-offset, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the offset in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b

M_POSITION_X +		Inquires the nominal search position for the X-coordinate. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies all X-coordinates.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the nominal search position, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the position in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y +		Inquires the nominal search position for the Y-coordinate. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies all Y-coordinates.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the nominal search position, in pixels, and can be specified with subpixel accuracy; for an M_SHAPE_... type of Model Finder context, specify the position in the units of the target's coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS +		Inquires the radius of the circle in the model. This is only valid for a model of type M_CIRCLE. This inquire type is not supported for a model defined in an M_SHAPE_ELLIPSE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value	Specifies the radius of the circle, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_ANGLE +		Inquires the reference axis angle of the model. By default, for an ellipse model defined in an M_SHAPE_ELLIPSE type of Model Finder context, the reference axis is aligned with the principal axis of the model (the width). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angle, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_X +		Inquires the X-coordinate, in pixels, of the reference axis origin of the model, relative to the model origin. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies that the default X-offset will be used. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-offset value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_REFERENCE_Y +		Inquires the Y-coordinate, in pixels, of the reference axis origin of the model, relative to the model origin. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies that the default Y-offset will be used. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-offset value. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_SAGITTA_TOLERANCE +		Inquires the tolerance for finding deformed circles (allowable radii variation) for an M_CIRCLE or an M_ELLIPSE type of model. This inquire type is only supported for M_CIRCLE or M_ELLIPSE types of models in an M_SHAPE_... type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 25.0 %.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the tolerance as a percentage of the allowable radii variation given the other Model Finder constraints. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SCALE +		Inquires the nominal search scale. For a model defined in an M_SHAPE_ELLIPSE or M_SHAPE_RECTANGLE type of Model Finder context, scale calculations are taken with respect to the model's width. For a model defined in an M_SHAPE_SEGMENT type of Model Finder context, scale calculations are taken with respect to the model's length. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the value of the nominal search scale. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SCALE_MAX_FACTOR +		Inquires the factor used to determine the upper limit (maximum permitted scale) of the scale range. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 2.0.	c		e	f	g	a		i		j	k	l	m	n	b

	M_INFINITE	Specifies that the upper limit of the scale range is infinite. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 <= *Value* <= 2.0	Specifies the factor that determines the maximum scale of the occurrence for a model in an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1.0	Specifies the factor that determines the maximum scale of the occurrence for a model in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_SCALE_MIN_FACTOR +		Inquires the factor used to determine the lower limit (minimum permitted scale) of the scale range. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.5.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 1.0	Specifies the factor that determines the minimum scale of the occurrence for a model in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	0.5 <= *Value* <= 1.0	Specifies the factor that determines the minimum scale of the occurrence for a model in an M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_ASPECT_RATIO_CONSTRAINT +		Inquires whether to constrain candidates to the nearest bound of the aspect ratio range if the candidate is outside of the range. This inquire type is only supported for a model in an M_SHAPE_ELLIPSE type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that for a candidate to be considered an occurrence, it must have an aspect ratio within the defined aspect ratio range. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to consider candidates with an aspect ratio that falls outside of the specified aspect ratio range, but constrain their fit to the closest bound of the aspect ratio range, and reduce their score accordingly. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SEGMENT_CONSISTENT_GRADIENT +		Inquires whether the gradients of the segment model occurrences need to be consistent. This inquire type is only supported for a model in an M_SHAPE_SEGMENT type of Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the segment model occurrences can have the same gradients or reverse gradients.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the gradients of the segment model occurrences must be consistent.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_LABEL +		Inquires the user-defined label. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_NO_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_LABEL	Specifies that no user label is associated with the model.	c		e	f	g	a		i		j	k	l	m	n	b

	Value	Specifies the user label of the model. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_VALID +		Inquires whether the model is valid or not. Note that a synthetic model is invalid if *size of the model box* * M_PIXEL_SCALE * M_SCALE is greater than 1024 pixels in the X-direction or the Y-direction. This inquire type is not supported for a model defined in an M_SHAPE_... type of Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the model is not valid.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the model is valid.	c		e	f	g	a		i		j	k	l	m	n	b
M_VERTICAL_THICKNESS +		Inquires the vertical thickness of the cross in the model. This is only valid for a model of type M_CROSS. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the vertical thickness of the cross, in model units.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIDTH +		Inquires the width of the shape in the model. This is only valid for a model of type M_CROSS, M_DIAMOND, M_ELLIPSE, M_RECTANGLE, M_SQUARE, and M_TRIANGLE. SET SET SET SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the width of the shape, in model units.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_CHAIN_ANGLE; M_CHAIN_INDEX; M_CHAIN_X; M_CHAIN_Y.

You can add the following value to the above-mentioned values to determine the required array size (number of elements) to store the returned values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining the required array size (number of elements) to store the returned values																	
InquireType combination value		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_NB_ELEMENTS		Retrieves the required array size (number of elements) to store the returned values.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

For model inquiries using the model's user label l(that is, when the Index parameter is set to the user label of a model), the InquireType parameter can be set to the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a model using its user label																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_INDEX_FROM_LABEL +		Inquires the model index associated with a model user label if the user label is used. Pass the user label as the Index parameter.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INVALID	Specifies that the user label is invalid. This is returned if the specified model label is not associated with a model.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the model index that is associated with the specified user label.	c		e	f	g	a		i		j	k	l	m	n	b

For result buffer inquiries, the InquireType parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a result buffer																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

	UserVarPtr - Possible values returned															
M_HAS_TARGET_EDGES_SAVED +		Inquires whether target edges have been saved. M_SAVE_TARGET_EDGES must be set to M_ENABLE to save target edges.														
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that target edges have not been saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that target edges have been saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOD_DEFINE_COMPATIBLE +		Inquires whether the information necessary to define a model from an occurrence will be saved in the result buffer. This inquire type is not supported for an M_SHAPE_... type of Model Finder result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to save any information.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to save all the necessary information.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_OUTPUT_UNITS +		Inquires whether results are returned in pixel or world units. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ACCORDING_TO_CALIBRATION.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ACCORDING_TO_CALIBRATION	Specifies that results are returned in world units if the result was calculated on an image associated with a camera calibration context; otherwise, specifies that results are returned in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL	Specifies that results are returned in pixel units, with respect to the pixel coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD	Specifies that results are returned in world units, with respect to the relative coordinate system. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_TYPE +		Inquires the type of result buffer allocated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GEOMETRIC	Specifies that the Model Finder result buffer can hold results from a search with an M_GEOMETRIC type of context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GEOMETRIC_CONTROLLED	Specifies that the Model Finder result buffer can hold results from a search with an M_GEOMETRIC_CONTROLLED type of context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHAPE_CIRCLE	Specifies that the Model Finder result buffer can hold results from a search with an M_SHAPE_CIRCLE type of context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHAPE_ELLIPSE	Specifies that the Model Finder result buffer can hold results from a search with an M_SHAPE_ELLIPSE type of context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHAPE_RECTANGLE	Specifies that the Model Finder result buffer can hold results from a search with an M_SHAPE_RECTANGLE type of context.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SHAPE_SEGMENT	Specifies that the Model Finder result buffer can hold results from a search with an M_SHAPE_SEGMENT type of context.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values listed in For inquiring about the number of occurrences to find for the entire context or an individual model specified using its index, For inquiring about the context; and for the following values: M_ACCEPTANCE; M_ACCEPTANCE_TARGET; M_ALLOC_OFFSET_X; M_ALLOC_OFFSET_Y; M_ALLOC_SIGN; M_ALLOC_SIZE_BAND; M_ALLOC_SIZE_BIT; M_ALLOC_SIZE_X; M_ALLOC_SIZE_Y; M_ALLOC_TYPE; M_ANGLE; M_ANGLE_DELTA_NEG; M_ANGLE_DELTA_POS; M_ASSOCIATED_CALIBRATION; M_BOX_MARGIN_BOTTOM; M_BOX_MARGIN_LEFT; M_BOX_MARGIN_RIGHT; M_BOX_MARGIN_TOP; M_BOX_OFFSET_X; M_BOX_OFFSET_Y; M_BOX_SIZE_X; M_BOX_SIZE_Y; M_CAD_Y_AXIS; M_CERTAINTY; M_CERTAINTY_TARGET; M_CORNER_RADIUS; M_COVERAGE_MAX; M_COVERAGE_MIN; M_FIT_ERROR_WEIGHTING_FACTOR; M_FIT_SCORE_MIN; M_FOREGROUND_VALUE; M_HAS_DONT_CARE_MASK; M_HAS_FLAT_REGIONS_MASK; M_HAS_WEIGHT_REGIONS_MASK; M_HEIGHT; M_HORIZONTAL_THICKNESS; M_INNER_RADIUS; M_LENGTH; M_MIN_SEPARATION_ANGLE; M_MIN_SEPARATION_ASPECT_RATIO; M_MIN_SEPARATION_SCALE; M_MIN_SEPARATION_X; M_MIN_SEPARATION_Y; M_MODEL_ASPECT_RATIO; M_MODEL_ASPECT_RATIO_MAX_FACTOR; M_MODEL_ASPECT_RATIO_MIN_FACTOR; M_MODEL_TYPE; M_ORIGINAL_X; M_ORIGINAL_Y; M_OUTER_RADIUS; M_PIXEL_SCALE; M_POLARITY; M_POSITION_DELTA_NEG_X; M_POSITION_DELTA_NEG_Y; M_POSITION_DELTA_POS_X; M_POSITION_DELTA_POS_Y; M_POSITION_X; M_POSITION_Y; M_RADIUS; M_REFERENCE_ANGLE; M_REFERENCE_X; M_REFERENCE_Y; M_SAGITTA_TOLERANCE; M_SCALE; M_SCALE_MAX_FACTOR; M_SCALE_MIN_FACTOR; M_SEARCH_ASPECT_RATIO_CONSTRAINT; M_USER_LABEL; M_VALID; M_VERTICAL_THICKNESS; M_WIDTH; M_RESULT_OUTPUT_UNITS.

You can add the following value to the above-mentioned values to determine the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the default value of an inquire type																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_DEFAULT		Inquires the default value of the specified inquire type.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for

Tables:

- For inquiring about the number of occurrences to find for the entire context or an individual model specified using its index
- For inquiring about the context or result buffer
- For inquiring about the context

- For inquiring about a model using its index
- For inquiring about a model using its user label
- For a result buffer

the values listed in all the parameter association tables.

You can add the following value to the above-mentioned values to determine whether an inquire type is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring whether an inquire type is supported																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_SUPPORTED		Inquires whether the specified inquire type is supported.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring about the number of occurrences to find for the entire context or an individual model specified using its index
- For inquiring about the context or result buffer
- For inquiring about the context
- For inquiring about a model using its index
- For inquiring about a model using its user label
- For a result buffer

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to the required data type.

> Note: In C++, when using a standard vector (std::vector) instead of an array, the values listed in this combination table must not be used. If you need to use a specific data type, instantiate the vector with the necessary data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type

InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
UserVarPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* **|** *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_ID	Casts the requested information to an *AIL_ID*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]* **|** *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]* **|** *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]* **|** *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]* **|** *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT*. If the requested information does not fit into an *AIL_INT*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodMask

Synopsis

Mask regions of a model.

Syntax

```
void MmodMask(
    AIL_ID ContextId,     //out
    AIL_INT Index,        //in
    AIL_ID MaskBufferId,  //in
    AIL_INT64 MaskType,   //in
    AIL_INT64 ControlFlag //in
)
```

Description

This function allows you to apply masks to a model. Each mask can set pixels in the specified model to one of three states: a "don't care" state, a "flat region" state, or a "weighted region" state. You can apply different types of masks to the same model by making multiple calls to this function. Note that "don't care" and "flat region" masks are not supported for synthetic models. In addition, this function does not support models in an M_SHAPE_... type of Model Finder context.

"Don't care" pixels in a model will not be considered when searching for occurrences of the model in the target.

"Flat region" pixels denote a region where no features are expected, other than consistent pixel values with no edges present; any edge found in a "flat region" reduces the target score.

"Weighted region" pixels denote that the extracted model features will have weights to indicate their importance in the occurrence. The weight of the features will affect the calculation of the score; although the weights themselves don't affect the target score, features corresponding to negative weights will be ignored when calculating the target score.

Note that if you modify the margin of a synthetic model's bounding box after a mask has been applied, the mask will be discarded since it is no longer the same size as the model.

Parameters

ContextId

Specifies the Model Finder context of the model to mask. For this parameter, you can only specify M_GEOMETRIC or M_GEOMETRIC_CONTROLLED type of Model Finder contexts. The Model Finder context must have been previously allocated on the system using MmodAlloc().

Index

Specifies the individual model for which to create a mask.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a model																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the index of the model for which the mask is being created. User labels cannot be used as indices; to retrieve the index of a model from its user label, use MmodInquire() with M_INDEX_FROM_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b

MaskBufferId

Specifies the identifier of the image buffer used to identify the masked pixels in the model. This buffer must be a 1-band 8-bit unsigned buffer for a "don't care" or "flat region" mask. For a "weighted region" mask, the buffer must be a 1-band 8-bit signed buffer. For M_GEOMETRIC and M_GEOMETRIC_CONTROLLED types of contexts, the buffer must be of the same size as the model. For M_SHAPE_... types of models, the buffer must be the same size as the smallest buffer in which you can draw the model box. You can inquire the model buffer size using MmodInquire() with M_ALLOC_SIZE_X and M_ALLOC_SIZE_Y. This buffer can be calibrated; however, even if the model is calibrated, the mask is applied on a pixel basis. Setting the MaskBufferId parameter to **M_NULL** when the MaskType parameter is set to M_DONT_CARE, M_FLAT_REGIONS, or M_WEIGHT_REGIONS removes the mask.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

MaskType

Specifies which type of mask to employ. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of mask																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DONT_CARE	Specifies that features present in the masked region (non-zero pixels) are ignored during the search. Features present in this region of the occurrence will not affect the scores.	c		e	f	g	a		i		j	k	l	m	n	b
M_FLAT_REGIONS	Specifies that no features are expected in the masked region (non-zero pixels). If features are present in this region of the occurrence, the target score will decrease.	c		e	f	g	a		i		j	k	l	m	n	b

M_WEIGHT_REGIONS	Specifies that the pixels corresponding to the model's active edges are weighted according to the values of the weighted region mask. The weights affect the calculation of the score; although the weights themselves don't affect the target score, edges corresponding to negative weights will be ignored when calculating the target score. The valid mask weights range from -127 to +127. Positive weights indicate how significant it is for a given pixel to be part of an edge in the occurrence; the more positive the weight, the greater the influence on the score. The absence of a pixel with a very positive weight will reduce the score more than the absence of one with a lower weight. The longer the edges with a positive weight, the more significant the edges to the target coverage and to the target score. Negative weights indicate how significant it is for a pixel not to be part of an edge in the occurrence; the more negative the weight, the greater the influence on the score. The presence of a pixel with a very negative weight will reduce the score more than the presence of one with a less negative weight. Note that weights that have no correspondence to edges in the model have no effect on the score.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodPreprocess

Synopsis

Preprocess a Model Finder context.

Syntax

```
void MmodPreprocess(
    AIL_ID ContextId,     //in-out
    AIL_INT64 ControlFlag //in
)
```

Description

This function preprocesses the specified Model Finder context. It extracts the active edges of the models contained within the Model Finder context and sets internal search settings so that future searches will be optimized for speed and robustness. The procedure is potentially quite lengthy (up to a few seconds) and increases the size of the Model Finder context significantly.

This function must be called before performing the first MmodFind() search. Call this function after all search settings have been set. When you save the Model Finder context, the model's preprocessing changes are not stored with the model. Upon restoration, the model must be preprocessed again.

Note that if some of the model's search settings are changed after a call to **MmodPreprocess**(), the model must be preprocessed again. To inquire if your model is in a preprocessed state, use MmodInquire() with M_PREPROCESSED (this value must be true to perform an MmodFind() search).

Parameters

ContextId

Specifies the Model Finder context of the model to preprocess. The Model Finder context must have been previously allocated on the system using MmodAlloc().

ControlFlag

Specifies whether to perform or undo the preprocessing of the model context. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether to perform preprocessing																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Performs the preprocessing of the model context.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESET	Undoes preprocessing. Undoing preprocessing can be useful if you want to conserve computer memory within an application and preserve Model Finder context settings.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodRestore

Synopsis

Restore a Model Finder context from disk.

Syntax

```
AIL_ID MmodRestore(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID SysId,                //in
    AIL_INT64 ControlFlag,       //in
    AIL_ID *ContextIdPtr         //out
)
```

Description

This function restores a Model Finder context that was previously saved to a file, using MmodSave() or MmodStream(). This function restores all of the Model Finder context's settings that were in effect when the Model Finder context was saved. A restored Model Finder context is not preprocessed, therefore you must call MmodPreprocess() before performing a search with MmodFind().

If you had associated camera calibration contexts with the models in your Model Finder context and you did not save them with the context, you must re-associate the camera calibration contexts, using MmodControl() with M_ASSOCIATED_CALIBRATION. Moreover, if you had previously set drawing control types, using MgraControl() with M_DRAW_..., then you must reset them since they were not saved with the context.

When the restored Model Finder context is no longer required, release it using MmodFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the restored Model Finder context's lifetime and you must not manually free it.

Parameters

FileName

Specifies the name and path of the file from which to restore the Model Finder context. The function handles (internally) the opening and closing of the file.

This parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Open** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b

AIL_TEXT("*FileName*") [1]	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, Model Finder context files have an MMF extension. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

SysId

Specifies the system on which to restore the Model Finder context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Specifies whether to restore, from the Model Finder context, the camera calibration contexts associated with the models.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether to restore the camera calibration contexts associated with the models																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the camera calibration contexts are not restored.	c		e	f	g	a		i		j	k	l	m	n	b

M_WITH_CALIBRATION	Specifies that the camera calibration contexts are restored. In this case, the camera calibration contexts must have been previously saved with the context, using MmodSave() or MmodStream() with M_WITH_CALIBRATION. Note that the camera calibration contexts associated with synthetic models are not saved or restored. The camera calibration information is restored from the same file as the Model Finder context. The camera calibration cannot be managed independently from the Model Finder context. When the Model Finder context is freed, the camera calibration is automatically freed as well.	c		e	f	g	a		i		j	k	l	m	n	b

ContextIdPtr

Specifies the address of the variable in which to write the Model Finder context identifier or specifies the data type that the function should use to return the Model Finder context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated Model Finder context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated Model Finder context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_MOD_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the Model Finder context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the restored Model Finder context. If restoration fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the Model Finder context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_MOD_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodSave

Synopsis

Save a Model Finder context to a file.

Syntax

```
void MmodSave(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID ContextId,            //in
    AIL_INT64 ControlFlag        //in
)
```

Description

This function saves all the information about the previously allocated Model Finder context to disk, including all of the individual models' current settings. This information can be reloaded, using MmodRestore() or MmodStream(). However, preprocessing and all drawing control type settings are not saved.

Parameters

FileName

Specifies the name and path of the file in which to save the Model Finder context; it is recommended that you use the MMF file extension for easier use with other Aurora Imaging Library software products. The function internally handles the opening and closing of this file. If this file already exists, it will be overwritten.

This parameter can be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Save As** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, Model Finder context files have an MMF extension. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

ContextId

Specifies the identifier of the Model Finder context to save.

ControlFlag

Specifies whether to save, with the Model Finder context, the camera calibration contexts associated with the models.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether to save the camera calibration contexts associated with the models																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the camera calibration contexts are not saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_WITH_CALIBRATION	Specifies that the camera calibration contexts are saved. The camera calibration information is saved in the same file as the Model Finder context. Note that the camera calibration contexts associated with synthetic models are not saved.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.

MmodStream

Synopsis

Load, restore, or save a Model Finder context from/to a file or a memory stream.

Syntax

```
void MmodStream(
    AIL_TEXT_PTR MemPtrOrFileName, //in-out
    AIL_ID SysId,                  //in
    AIL_INT64 Operation,           //in
    AIL_INT64 StreamType,          //in
    AIL_DOUBLE Version,            //in
    AIL_INT64 ControlFlag,         //in
    AIL_ID *ContextModIdPtr,       //in-out
    AIL_INT *SizeByteVarPtr        //out
)
```

Description

This function can load, restore, or save a Model Finder context from/to a file or memory stream.

To inquire the number of bytes necessary to save a Model Finder context to memory stream, you should first call this function (**MmodStream**()) with M_INQUIRE_SIZE_BYTE.

The content saved to memory stream is equivalent to the content saved to file. This function is equivalent to a file saved using MmodSave().

You can use this and other Aurora Imaging Library stream functions, for example, to save all required Aurora Imaging Library objects, as well as any other custom data, for your application to a memory stream. Once in a memory stream, you can write the stream to a single file or transfer it over a network. You are responsible for concatenating the streams and for saving the stream to file. All information about the previously allocated Model Finder context is saved, including all of the individual models' current settings. However, preprocessing, any associated camera calibration contexts and drawing control type settings are not saved.

All of the Model Finder context's settings that were in effect when the Model Finder context was saved will be restored. A loaded or restored Model Finder context is not preprocessed, therefore you must call MmodPreprocess() before performing a search with MmodFind(). If you had associated camera calibration contexts with the models in your Model Finder context and you did not save them with the context, you must re-associate the camera calibration contexts, using MmodControl() with M_ASSOCIATED_CALIBRATION.

Using **MmodStream**(), you can choose to save a backwards-compatible version of the Model Finder context, which will work using a version of Aurora Imaging Library that is up to one major release older than the current version (depending on which version is specified). However, all settings and features unique to the higher version will be ignored when restored using the lower version. Besides saving backwards-compatible versions, you can also load or restore Model Finder contexts saved using Aurora Imaging Library version 7.0 or above. Settings that do not exist in the lower version will be filled with default values when the Model Finder context is loaded or restored.

Parameters

MemPtrOrFileName
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_TEXT_PTR [optionally, in C++: a reference to an AIL_STRING]
- Optionally, in C++: a reference to a std::vector of type AIL_UINT8

Specifies the file or memory stream.

See the Parameter associations section for possible values that can be returned/specified.

SysId

Specifies the system on which to restore the Model Finder context.

See the Parameter associations section for possible values that can be specified.

Operation

Specifies the operation to perform on the Model Finder context.

See the Parameter associations section for possible values that can be specified.

StreamType

Specifies the type of stream in which to store/from which to restore the Model Finder context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of stream																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILE	Specifies a file stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_MEMORY	Specifies a memory stream. You are responsible for allocating a block of memory for the stream.	c		e	f	g	a		i		j	k	l	m	n	b

Version

Specifies the Aurora Imaging Library version of the Model Finder context. When performing an M_LOAD or M_RESTORE operation, this parameter must be set to **M_DEFAULT**.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Specifies whether to restore, from the file or memory stream, the camera calibration contexts associated with the models, or specifies to save, with the Model Finder context, the camera calibration contexts associated with the models.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether to restore or save the camera calibration contexts																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies that the camera calibration contexts are not saved or restored.	c		e	f	g	a		i		j	k	l	m	n	b
M_WITH_CALIBRATION	Specifies that the camera calibration contexts are saved or restored, depending on the Operation parameter. When restored, the camera calibration contexts must have been previously saved with the context, using MmodSave() or **MmodStream**() with M_WITH_CALIBRATION. Note that the camera calibration contexts associated with synthetic models are not saved or restored. The camera calibration information is restored from/ saved to the same file or memory stream as the Model Finder context. The camera calibration cannot be managed independently from the Model Finder context. When the Model Finder context is freed, the camera calibration is automatically freed as well.	c		e	f	g	a		i		j	k	l	m	n	b

ContextModIdPtr

Specifies the address of the variable in which to write or from which to read the identifier of the Model Finder context.

See the Parameter associations section for possible values that can be returned/specified.

SizeByteVarPtr

Specifies the address of the variable in which to write the size of the Model Finder context, in bytes.

If the size is not required, you can set this parameter to **M_NULL**.

> Note that the size of a Model Finder context will vary depending on the Aurora Imaging Library version specified.

Parameter associations

The table below lists possible values for the **MemPtrOrFileName**, **SysId**, **Operation**, **Version**, and **ContextModIdPtr** parameters.

- For performing the stream operation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For performing the stream operation.																	
Operation		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	MemPtrOrFileName - Possible values passed or returned																
	SysId																
	Version																

ContextModIdPtr - Possible values passed or returned																
M_INQUIRE_SIZE_BYTE	Inquires the number of bytes required to save a Model Finder context to memory stream. This operation is not supported when the StreamType parameter is set to M_FILE.															
MemPtrOrFileName	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the Model Finder context must be compatible when saved. The size of a Model Finder context might differ from one Aurora Imaging Library version to another.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP4	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 4.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP5	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 5.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
ContextModIdPtr	Specifies the address of the variable containing the identifier of the Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOAD	Loads the content of a specified file or memory stream into a previously allocated Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b

MemPtrOrFileName	Specifies the file or memory stream from which to load the context.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. Model Finder contexts typically have an MMF file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b

ContextModIdPtr	Specifies the address of the variable containing the identifier of the Model Finder context. All previous settings are overwritten with those of the loaded context, or are set to the default value if the settings didn't exist in the version of Aurora Imaging Library from which the context is being loaded.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESTORE	Restores a Model Finder context from a file or memory stream and assigns it an Aurora Imaging Library identifier.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream from which to restore the context.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. Model Finder contexts typically have an MMF file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b

SysId	Specifies the system on which to allocate the Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
ContextModIdPtr	Specifies the address of the variable in which to write the identifier of the Model Finder context. If the operation is not successful, **M_NULL** is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE	Saves a Model Finder context to a specified file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. For easier use with other Aurora Imaging Library software products, when saving a model finder context to a file, use the MMF file extension. The function handles (internally) the opening and closing of the file. If the file already exists, it will be overwritten. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

		MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which to write, and the block of memory must be large enough to stream the entire object. To determine the required size, call this function with M_INQUIRE_SIZE_BYTE. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
	SysId		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
	Version		Specifies the Aurora Imaging Library version with which the Model Finder context must be compatible when saved. Note that the Model Finder context can only be up to one major release older than the current version.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP4	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 4.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP5	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 5.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
	ContextModIdPtr		Specifies the address of the variable containing the identifier of the Model Finder context to save.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailmod11.lib.
DLL	Requires ail11.dll; ailmod11.dll.